

Unconditional all-orders asymptotics for two OEIS permutation sequences

Stable path forests, Poisson expansions, and index inversion

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Abstract

The OEIS entries A189281 and A110128 count permutations avoiding a prescribed signed or absolute difference between entries two positions apart. Their displayed higher asymptotic expansions are associated with conjectured recurrences. We give a direct, recurrence-free proof of those expansions, together with a finite exact formula for every coefficient. More generally, for any fixed positive offsets r, s , the complete probability generating function of the number of violations has an all-orders expansion consisting of $e^{\theta u}$ times polynomial corrections, where $\theta = 1$ in the oriented case and $\theta = 2$ in the absolute case. The remainder is uniform on compact complex u -sets.

The structural ingredient is an exact stabilization theorem for tilings of long path forests: low factorial moments depend on the numbers of paths, not their individual lengths. We also prove a quantitative total-variation universality bound, a factorial denominator bound for all asymptotic coefficients, and an all-orders Lambert- W index inversion. Exact rational computations recover the OEIS coefficients through order ten and extend both expansions through order sixteen. Independent permutation enumeration and Bonferroni certificates accompany the proofs. The specific conjectured recurrences and the stronger all-orders integrality suggested by A189281 are not assumed or claimed proved. A concrete rational-function conjecture is proposed as a route to the latter question.

Status. This is a proof-focused research draft, not a claim of peer review or a Lean-checked formalization. The theorems below have mathematical proofs; the computations are reproducible exact checks. Historical priority beyond the sources reviewed here remains to be assessed. The results establish the algebraic asymptotic sector, not a complete exponentially improved transseries.

Editorial note (1 October 2026, batch 73O2). This report is manuscript 60 of batch 73 of ProveIt’s incoming-reports intake, placed in the research-report collection in commit [6e193dd4f](#). It is AI-assisted, unrefereed and not formalized. The text is the delivered manuscript; the write added only the label prefix `spf:`, the bracketed notes marked “[Added 1 October 2026, batch 73O2]”, Remarks [7.2](#) and [9.1](#), three bibliography entries, and [section B](#), which records the provenance and every editorial change. The rational-collapse statement of [section 11](#) remains a conjecture.

Editorial note (1 October 2026, batch 74). Since batch 74 the report has two Parts. Part I is manuscript 60, as described above. Part II ([section 14](#)) adds manuscript 03 of batch 74, an independent second derivation of Part I’s main theorems, and prints only what it adds: uniformity over forest shapes, comparison bounds, closed forms of $c_3(r, s)$, and proved integrality for the oriented one-path family. Its labels carry the prefix `fga:`.

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Part I

Unconditional all-orders asymptotics for two OEIS permutation sequences

1 The target and the precise contribution

For a uniformly random $\pi \in S_n$, define, for fixed $r, s \geq 1$,

$$X_{r,s,n}^{(1)} = \sum_{i=1}^{n-r} \mathbf{1}\{\pi(i+r) - \pi(i) = s\}, \quad (1.1)$$

$$X_{r,s,n}^{(2)} = \sum_{i=1}^{n-r} \mathbf{1}\{|\pi(i+r) - \pi(i)| = s\}. \quad (1.2)$$

An empty sum is zero. Write

$$A_{r,s}^{(\theta)}(n) = n! \mathbb{P}(X_{r,s,n}^{(\theta)} = 0), \quad \theta \in \{1, 2\}. \quad (1.3)$$

In this notation,

$$A189281(n) = A_{2,2}^{(1)}(n), \quad A110128(n) = A_{2,2}^{(2)}(n). \quad (1.4)$$

These identifications and the sequence definitions are recorded in the OEIS entries [1, 2].

1.1 What the sources do, and do not, already establish

The A189281 entry lists an order-eight, degree-eleven recurrence as conjectural, and describes its displayed terms through n^{-10} as consequences of that recurrence. It also records independent dynamic-programming verification of sequence values through $n = 300$ [1]. The recurrence-guessing work of Kauers and Koutschan is an important source for this problem [5]. For A110128, the entry lists a guessed order-24, degree-64 operator and attributes its higher asymptotic terms to that operator [2].

The common leading limits are not new. Tauraso proved, in the diagonal absolute case, the first correction

$$\frac{A_{r,r}^{(2)}(n)}{n!} = e^{-2} \left(1 + \frac{4(r-1)}{n} + O(n^{-2}) \right) \quad (1.5)$$

and gave exact counting formulas [4]. Spahn and Zeilberger developed an exact matching-of-tilings formula for both offset families, explained its computational use, and discussed Matsuo's alternative approach [3]. We use their combinatorial framework, with a self-contained proof, rather than claiming the underlying inclusion-exclusion formula as new.

There is also a relevant recent development. Padhi's August 2026 preprint reports holonomicity for every fixed r, s , while explicitly recording only a partial result toward the specific A189281 operator [6]. Our arguments neither assume that preprint's holonomicity theorem nor prove a particular recurrence. A general existence theorem for a recurrence and a proof of a specific guessed operator are distinct questions. [Added 1 October 2026, batch 73O2: the intake confirmed that the arXiv record of [6] exists under the title given in the bibliography, but did not check its content. Nothing below depends on it.]

1.2 The gap addressed here

We supply an unconditional route from the counting problem to every coefficient and a controlled remainder. The important point is not merely to fit more terms. One must justify replacing

an n -dependent inclusion-exclusion sum by an infinite coefficient calculation, including the contributions of arbitrarily many isolated selected edges. The exact stabilization theorem and a uniform factorial-moment bound supply that justification.

The resulting contribution has four connected parts. First, the coefficients attached to the two conjectural recurrences become theorems about the original permutation counts. Second, the same proof gives every fixed pair of offsets and the full violation distribution. Third, it explains why congruence classes of n do not enter any algebraic-order coefficient. Fourth, it gives an inverse expansion with a precise interpretation for a discrete sequence.

The ProveIt repository was consulted as the project context. Its Fabius asymptotic audit carefully separates a printed approximation, a necessary correction, and an all-orders theorem [9]. Here the analogous separation is between finite data, formal coefficient extraction, a uniform asymptotic theorem, and an unproved stronger closed form. No declaration from the repository is a mathematical dependency of the present proof, and no new repository build is reported.

1.3 A preview of the main theorem

Put

$$G_{r,s,n}^{(\theta)}(u) = \mathbb{E}(1+u)^{X_{r,s,n}^{(\theta)}}. \quad (1.6)$$

The use of $1+u$, rather than u , makes the coefficients factorial moments.

Theorem 1.1 (All-orders polynomial correction theorem). *For fixed positive integers r, s , $\theta \in \{1, 2\}$, every integer $M \geq 0$, and every $0 \leq U < \infty$, there are polynomials $B_J^{(\theta)}(u; r, s) \in \mathbb{Q}[u]$ such that, uniformly for $|u| \leq U$,*

$$G_{r,s,n}^{(\theta)}(u) = e^{\theta u} \sum_{J=0}^M \frac{B_J^{(\theta)}(u; r, s)}{n^J} + O_{M,r,s,U}(n^{-M-1}). \quad (1.7)$$

They are given by the finite formula (5.11). They satisfy

$$B_0 = 1, \quad \deg B_J \leq 2J, \quad B_J(0) = 0 \quad (J \geq 1), \quad (1.8)$$

$$B_J(u; r, s) = B_J(u; s, r), \quad (2J)!B_J(u; r, s) \in \mathbb{Z}[u], \quad (1.9)$$

$$B_1^{(\theta)}(u; r, s) = \theta(1-r-s)u + \theta(1-\theta)u^2. \quad (1.10)$$

In particular, for $c_J^{(\theta)}(r, s) = B_J^{(\theta)}(-1; r, s)$,

$$\frac{A_{r,s}^{(\theta)}(n)}{n!} = e^{-\theta} \left(\sum_{J=0}^M \frac{c_J^{(\theta)}(r, s)}{n^J} + O(n^{-M-1}) \right), \quad c_0 = 1, \quad c_1 = \theta(r+s-\theta). \quad (1.11)$$

The proof occupies sections 2 to 6. The arithmetic assertions are proved in section 7.

2 Selected violations and pairs of path tilings

2.1 Why paths are the right objects

The directed graph on $\{1, \dots, n\}$ with edges $i \rightarrow i+r$ is a disjoint union of r directed paths, when $n \geq r$. Their vertex counts are $\lfloor n/r \rfloor$ and $\lceil n/r \rceil$. The value graph with edges $j \rightarrow j+s$ is similarly a union of s paths.

More generally, consider any directed path forest on the position vertices and any directed path forest on the value vertices, both having n vertices. The oriented statistic counts position edges mapped to directed value edges by a uniform bijection. The absolute statistic counts position edges mapped to value edges in either direction. The proofs below apply to this more general model as well.

Selecting a subset of the position edges gives a collection of nontrivial path components and singleton vertices. For the oriented statistic, each selected component must map in its prescribed order to a value-path interval. For the absolute statistic there are exactly two choices of direction per nontrivial component. A reversal of direction *within* one component is impossible: at the first change of sign it would repeat the preceding value. Thus the orientation weight is 2^c , not 2^k , where c is the number of components and k the number of selected edges.

Definition 2.1 (Component profile). A profile is a finitely supported sequence $\mathbf{b} = (b_2, b_3, \dots)$ of nonnegative integers. Define

$$\begin{aligned} c &= \sum_{\ell \geq 2} b_\ell, & k &= \sum_{\ell \geq 2} (\ell - 1)b_\ell, \\ v &= \sum_{\ell \geq 2} \ell b_\ell = k + c, & d &= k - c = \sum_{\ell \geq 2} (\ell - 2)b_\ell, \end{aligned} \quad (2.1)$$

and $\mathbf{b}! = \prod_{\ell \geq 2} b_\ell!$. The number d is the *defect*: dimers have defect zero, and each extra edge in a component adds one unit of defect.

For a position forest $\mathcal{P}$, let $C_{\mathcal{P}}(\mathbf{b})$ be its number of tilings with b_ℓ components of size $\ell \geq 2$, and $n - v$ singletons. A tiling is simply a partition into consecutive intervals along each path; unused edges may lie between neighboring intervals.

Proposition 2.2 (Exact profile identity). *For position and value forests $\mathcal{P}, \mathcal{V}$ on n vertices,*

$$\mathbb{E}(1 + u)^X = \sum_{\mathbf{b}: v \leq n} \theta^c u^k C_{\mathcal{P}}(\mathbf{b}) C_{\mathcal{V}}(\mathbf{b}) \frac{(n - v)! \mathbf{b}!}{n!}. \quad (2.2)$$

Infeasible profiles have zero tiling count.

Proof. For each permutation, expand $(1 + u)^X$ as a sum over its selected violation sets. Such a set determines its input tiling and its image tiling, with the same component profile. Match the b_ℓ input components of each length to the b_ℓ output components in $b_\ell!$ ways. Choose a direction for each nontrivial component when $\theta = 2$, and match the $n - v$ singletons arbitrarily. This gives exactly $\theta^c (n - v)! \mathbf{b}!$ permutations for each pair of tilings, counted with the selected set distinguished. Division by $n!$ proves the identity.

Additional violations outside the selected set are allowed. This is essential: the identity counts marked subsets of violations, not permutations having exactly that selected set. $\square$

At $u = -1$, (2.2) is the matching-of-tilings inclusion-exclusion formula of Spahn and Zeilberger [3]. Its extension to general u makes the later distributional statements possible.

3 Exact stabilization on long path forests

Write $x^{\underline{j}} = x(x - 1) \cdots (x - j + 1)$ and $x^{\bar{j}} = x(x + 1) \cdots (x + j - 1)$, with both empty products equal to one. For a profile $\mathbf{b}$, put

$$E_h(\mathbf{b}) = [z^h] \prod_{\ell \geq 2} (1 + (\ell - 1)z)^{b_\ell}. \quad (3.1)$$

Define an integer polynomial

$$F_r(n; \mathbf{b}) = \sum_{h=0}^c (-1)^h (r - 1)^{\bar{h}} E_h(\mathbf{b}) (n - k - h)^{\underline{c-h}}. \quad (3.2)$$

Its degree in n is c and its leading coefficient is one. For $r = 1$, it reduces to $(n - k)^{\underline{c}}$.

Theorem 3.1 (Stable tiling polynomial). *Let $\mathcal{P}$ be a union of r directed paths with vertex counts $L_1, \dots, L_r$ and total n . If each $L_i \geq k$, then*

$$C_{\mathcal{P}}(\mathbf{b}) = \frac{F_r(n; \mathbf{b})}{\mathbf{b}!}. \quad (3.3)$$

Thus, in this range, the answer depends on $n, r, \mathbf{b}$, but not on the individual path lengths.

3.1 A formal coefficient identity

The following residue calculation is included to make the stabilization mechanism explicit. Only finitely many variables are needed for any specified coefficient.

Let $z_2, z_3, \dots$ be formal variables and let W , with zero constant term, solve

$$W = \sum_{\ell \geq 2} z_{\ell} (1 + W)^{-(\ell-1)}. \quad (3.4)$$

Define

$$\mathcal{D} = 1 + \sum_{\ell \geq 2} (\ell - 1) z_{\ell} (1 + W)^{-\ell}. \quad (3.5)$$

For every integer N and every profile,

$$[\mathbf{z}^{\mathbf{b}}] \frac{(1 + W)^N}{\mathcal{D}} = \frac{(N - k)^c}{\mathbf{b}!}. \quad (3.6)$$

To prove this, replace every z_{ℓ} by tz_{ℓ} and put $\phi(w) = \sum z_{\ell} (1 + w)^{-(\ell-1)}$. Then $W = t\phi(W)$ and $\mathcal{D} = 1 - t\phi'(W)$. Work first over the fraction field in finitely many z_{ℓ} , so that $\phi(0)$ is invertible; the resulting identity is polynomial coefficientwise. The change of formal residue variable $t = w/\phi(w)$ gives

$$[t^c] \frac{f(W)}{1 - t\phi'(W)} = [w^c] f(w) \phi(w)^c. \quad (3.7)$$

Indeed, the Jacobian factor cancels $1 - t\phi'(W)$. The coefficient of $\mathbf{z}^{\mathbf{b}}$ in $\phi(w)^c$ is $c!(1 + w)^{-k}/\mathbf{b}!$. Taking $f(w) = (1 + w)^N$ proves (3.6).

3.2 Proof of the stabilization theorem

For a single path of length L , a profile with $v \leq L$ has $L - k$ tiles in all. Ordering those tiles gives

$$C_L(\mathbf{b}) = \frac{(L - k)!}{(L - v)! \mathbf{b}!} = \frac{(L - k)^c}{\mathbf{b}!}. \quad (3.8)$$

If $k \leq L < v$, the falling factorial is zero, again giving the correct count. Consequently, through weighted edge degree $k \leq L$, the actual tiling series agrees with the formal surrogate $(1 + W)^L/\mathcal{D}$.

For r paths, all allocations of a total profile of edge degree k to the paths have local edge degree at most k . Under $L_i \geq k$, multiplying the surrogates is therefore legitimate at the required coefficient. Their product is

$$\frac{(1 + W)^n}{\mathcal{D}^r}. \quad (3.9)$$

Expand the additional factor $\mathcal{D}^{-(r-1)}$. A selected subprofile $\mathbf{a} \leq \mathbf{b}$, with $|\mathbf{a}| = h$, contributes

$$\frac{(-1)^h (r - 1)^{\bar{h}}}{\mathbf{a}!} \prod_{\ell \geq 2} (\ell - 1)^{a_{\ell}} \mathbf{z}^{\mathbf{a}} \frac{(1 + W)^{n - \sum \ell a_{\ell}}}{\mathcal{D}}. \quad (3.10)$$

Applying (3.6) to the remaining profile gives

$$\frac{(-1)^h (r-1)^{\bar{h}} \prod (\ell-1)^{a_\ell}}{\mathbf{a}!(\mathbf{b}-\mathbf{a})!} (n-k-h)^{\underline{c-h}}. \quad (3.11)$$

Summing over subprofiles of size h replaces the product of binomial coefficients by $E_h(\mathbf{b})/\mathbf{b}!$. This is exactly (3.2)–(3.3), proving theorem 3.1. $\square$

Corollary 3.2 (Stable factorial moments). *If the position forest has r paths, the value forest has s paths, every path has at least k vertices, and $2k \leq n$, then*

$$\mu_k := \mathbb{E} \binom{X}{k} = \sum_{\mathbf{b}: \sum (\ell-1)b_\ell = k} \frac{\theta^c F_r(n; \mathbf{b}) F_s(n; \mathbf{b})}{\mathbf{b}! n^v}. \quad (3.12)$$

For the ordinary offset model, this is exact whenever

$$k \leq \min \left(\left\lfloor \frac{n}{r} \right\rfloor, \left\lfloor \frac{n}{s} \right\rfloor, \left\lfloor \frac{n}{2} \right\rfloor \right). \quad (3.13)$$

Proof. Use theorem 3.1 on both tiling counts in (2.2) and extract u^k . The condition $2k \leq n$ ensures $v = k + c \leq n$ for every profile in the displayed sum. $\square$

As a first check, a single dimer has $F_r = n - r$, so, for $n \geq \max(r, s, 2)$,

$$\mathbb{E} X = \frac{\theta(n-r)(n-s)}{n(n-1)}. \quad (3.14)$$

The same formula follows directly by counting the available input and output edges.

4 Uniform bounds and beyond-all-orders universality

An exact formula for fixed moments is not, by itself, enough to justify an asymptotic expansion of a probability. A bound uniform in the moment order is needed.

Lemma 4.1 (Uniform factorial-moment bound). *For either statistic and arbitrary path forests on $n \geq 1$ vertices,*

$$0 \leq \mu_k \leq \frac{(\theta e^2)^k}{k!} \quad (k \geq 0). \quad (4.1)$$

Consequently, for $|u| \leq U$ and $a = \theta e^2 U$,

$$\left| G(u) - \sum_{k=0}^K \mu_k u^k \right| \leq e^a \frac{a^{K+1}}{(K+1)!}. \quad (4.2)$$

Proof. Fix a selected set of k position edges, with c components and $v = k + c$ vertices. There are at most $\theta^c n^c$ ways to assign starts and directions to its components in the value forest, even before excluding overlaps. Hence the probability that all the selected edges are violations is at most $\theta^c n^c / n^v$.

For $0 \leq v \leq n$, the geometric mean of the largest v factors in $n!$ is at least the geometric mean of all n factors. Also $n! \geq (n/e)^n$. Thus $n^v \geq (n/e)^v$, and the last probability is at most $\theta^k e^{2k} n^{-k}$, since $c \leq k$ and $v \leq 2k$. There are at most $n^k/k!$ possible selected edge sets. Summing proves (4.1). Summing its exponential-series tail proves (4.2). For $k = 0$ the assertion is immediate. $\square$

Theorem 4.2 (Quantitative independence of long path lengths). *Consider two position-value forest pairs, each on n vertices, with the same numbers r and s of paths. Suppose every path in both pairs has at least K vertices and $2K \leq n$. Let X and $\tilde{X}$ be the corresponding statistics for the same θ . For every $R > 1$, with $a = \theta e^2(R + 1)$,*

$$d_{\text{TV}}(\mathcal{L}(X), \mathcal{L}(\tilde{X})) \leq \frac{R}{R-1} e^a \frac{a^{K+1}}{(K+1)!}. \quad (4.3)$$

If all path lengths are at least ηn for a fixed $\eta > 0$, the right-hand side can be chosen to be $\exp(-\eta' n \log n + O(n))$ for every fixed $0 < \eta' < \min(\eta, 1/2)$.

Proof. By Corollary 3.2, the two sets of factorial moments agree through order K . On the circle $|t| = R$, use $u = t - 1$ in (4.2); the two ordinary probability generating functions differ by at most $2e^a a^{K+1}/(K+1)!$. Cauchy's coefficient inequality bounds the difference of the probabilities at j by this quantity times R^{-j} . Sum over $j \geq 0$ and divide by two to obtain (4.3). Finally, take $K = \lfloor \eta' n \rfloor$ and use $\log(K!) = K \log K - K + O(\log K)$. $\square$

This statement is stronger than agreement of a few asymptotic coefficients. It concerns the entire probability law. It also clarifies a limitation: the bound does not identify the leading exponentially small difference between two particular forest geometries. That finer sector is left open.

5 A finite formula for every asymptotic coefficient

5.1 Normalization and the defect filtration

Put $x = 1/n$ and define

$$f_{r,\mathbf{b}}(x) = \sum_{h=0}^c (-1)^h (r-1)^{\bar{h}} E_h(\mathbf{b}) x^h \prod_{a=0}^{c-h-1} (1 - (k+h+a)x), \quad (5.1)$$

$$D_v(x) = \prod_{a=0}^{v-1} (1 - ax), \quad (5.2)$$

$$Q_{\mathbf{b}}(x) = \frac{f_{r,\mathbf{b}}(x) f_{s,\mathbf{b}}(x)}{D_v(x)}. \quad (5.3)$$

Then $f_{r,\mathbf{b}}(1/n) = n^{-c} F_r(n; \mathbf{b})$, and a stabilized profile contributes

$$\frac{\theta^c u^k}{\mathbf{b}!} n^{-d} Q_{\mathbf{b}}(1/n) \quad (5.4)$$

to the generating function. This displays why defect $d > J$ cannot contribute to the coefficient of n^{-J} .

Separate the dimers from the other components:

$$m = b_2, \quad \beta = (b_3, b_4, \dots). \quad (5.5)$$

For the latter profile write

$$C = \sum_{\ell \geq 3} \beta_\ell, \quad K = \sum_{\ell \geq 3} (\ell - 1) \beta_\ell, \quad d = K - C, \quad \beta! = \prod_{\ell \geq 3} \beta_\ell!. \quad (5.6)$$

Thus $c = m + C$, $k = m + K$, $v = 2m + K + C$. The profiles β of a given defect d are in bijection with ordinary integer partitions of d : a component of length ℓ contributes a part $\ell - 2$.

Lemma 5.1 (Polynomial dependence on the number of dimers). *For fixed β, j, r, s ,*

$$P_{\beta,j}(m) := [x^j]Q_{(m,\beta)}(x) \quad (5.7)$$

is a polynomial in m over $\mathbb{Q}$ of degree at most $2j$. For positive integer r, s , it takes integer values at every $m \geq 0$.

Proof. The coefficient $E_h(m, \beta)$ is a polynomial in m of degree at most h , since the dimer contribution is $(1+z)^m$. For a product of consecutive linear factors

$$\prod_{a=0}^{m+C-h-1} (1 - (m + K + h + a)x), \quad (5.8)$$

the coefficient of x^q is polynomial in m of degree at most $2q$. One way to see the bound is to take its formal logarithm. The coefficient of x^i is a constant multiple of a sum of i th powers over a consecutive interval with affine endpoints, and therefore has degree at most $i + 1$. Exponentiating shows degree at most $2q$, because a product of power sums of total x -degree q has degree at most $q + q$. The same argument applies to $D_v(x)^{-1}$.

In a term carrying the additional factor $E_h x^h$, the degree at coefficient x^j is at most $h + 2(j - h) \leq 2j$. Products preserve the total bound. For the few small nonnegative m for which $c < h$, $E_h = 0$; thus the polynomial extension gives the same zero contribution. Equivalently one can define the consecutive power sums by their polynomial extensions before multiplying by E_h .

Finally, for nonnegative m , the numerator polynomials in (5.3) have integer coefficients and $D_v^{-1} \in \mathbb{Z}[[x]]$. Hence every value in (5.7) is an integer. $\square$

5.2 Poisson summation by finite differences

Let $\Delta P(m) = P(m+1) - P(m)$. For a polynomial of degree at most q ,

$$P(m) = \sum_{t=0}^q \frac{\Delta^t P(0)}{t!} m^t. \quad (5.9)$$

Therefore the elementary identity

$$\sum_{m \geq 0} \frac{z^m}{m!} P(m) = e^z \sum_{t=0}^q \frac{\Delta^t P(0)}{t!} z^t \quad (5.10)$$

turns an infinite dimer sum into a finite polynomial. It follows either by summing each falling factorial, or by differentiating e^z exactly t times.

Definition 5.2 (Coefficient formula). For $J \geq 0$, define

$$B_J^{(\theta)}(u; r, s) = \sum_{d=0}^J \sum_{\substack{\beta \geq 0 \\ \sum_{\ell \geq 3} (\ell-2)\beta_\ell = d}} \frac{\theta^C u^K}{\beta!} \sum_{t=0}^{2(J-d)} \frac{\Delta^t P_{\beta, J-d}(0)}{t!} (\theta u)^t. \quad (5.11)$$

All sums on the right are finite.

Only evaluations at $m = 0, 1, \dots, 2(J-d)$ are required. Their sufficiency is a proved degree bound, not a polynomial guess. No term $A(n)$ and no recurrence for $A(n)$ is needed to compute B_J .

6 Proof of the uniform all-orders expansion

We now justify the asymptotic interpretation of (5.11). This is the analytic step that a purely formal expansion of the profile formula would leave unresolved.

Lemma 6.1 (Uniform analytic bound for a profile). *For every profile of edge degree k , put $\rho_k = 1/[8(k+1)^2]$. There is a constant $C_{r,s}$, independent of the profile and k , such that*

$$\sup_{|x| \leq \rho_k} |Q_{\mathbf{b}}(x)| \leq C_{r,s}. \quad (6.1)$$

One may take $C_{r,s} = e^{2^{r+s}}$. In particular,

$$|[x^j]Q_{\mathbf{b}}(x)| \leq C_{r,s}[8(k+1)^2]^j, \quad (6.2)$$

and, if $n \geq 16(k+1)^2$,

$$\left| Q_{\mathbf{b}}(1/n) - \sum_{j=0}^q [x^j]Q_{\mathbf{b}}(x)n^{-j} \right| \leq 2C_{r,s} \left(\frac{8(k+1)^2}{n} \right)^{q+1}. \quad (6.3)$$

Proof. The elementary symmetric function bound $E_h(\mathbf{b}) \leq k^h/h!$ follows by expanding the h th power of the sum of the component edge lengths. Every integer appearing in a numerator factor is at most $2k$, and there are at most k such factors. Thus, on the stated disk,

$$|f_{r,\mathbf{b}}(x)| \leq e^{2k^2\rho_k} \sum_{h \geq 0} \frac{(r-1)^{\bar{h}}}{h!} (k\rho_k)^h = e^{2k^2\rho_k} (1 - k\rho_k)^{-(r-1)}. \quad (6.4)$$

For $r = 1$, the last sum is simply one. Also $v \leq 2k$ and $a\rho_k < 1/2$ for $0 \leq a < v$, so

$$|D_v(x)^{-1}| \leq \exp \left(2\rho_k \sum_{a=0}^{v-1} a \right) \leq e^{4k^2\rho_k}. \quad (6.5)$$

Combining these bounds gives (6.1) with the stated generous constant. The denominator has no zero on the disk. Cauchy's coefficient bound then proves (6.2), and its geometric tail at $1/n \leq \rho_k/2$ proves (6.3). $\square$

Proof of theorem 1.1: analytic assertion. Fix M, r, s, U , and for sufficiently large n choose

$$T_n = \left\lfloor \frac{(2M+4) \log n}{\log \log n} \right\rfloor. \quad (6.6)$$

The factorial-moment tail (4.2) with $K = T_n$ is $n^{-(2M+4)+o(1)}$, hence is $O(n^{-M-1})$. For fixed offsets, every position and value path has more than T_n vertices eventually. Also $2T_n \leq n$ and $16(T_n+1)^2 \leq n$. Thus all retained moments are given exactly by (3.12), and Lemma 6.1 applies to every retained profile.

High defect. The sum of absolute profile weights, marked by their defect, has the convergent generating function

$$\sum_{\mathbf{b}} \frac{\theta^c U^k}{\mathbf{b}!} z^d = \exp \left(\sum_{q \geq 1} \theta U^q z^{q-1} \right) = \exp \left(\frac{\theta U}{1 - Uz} \right). \quad (6.7)$$

For $U = 0$ the assertion is immediate; otherwise use a disk strictly inside $|z| < 1/U$. Its Taylor tail of degree M at $z = 1/n$ is $O(n^{-M-1})$. Together with (6.1), this bounds all retained profiles of defect $d \geq M+1$ by $O(n^{-M-1})$.

Low defect. For each $d \leq M$, there are only finitely many β . Expand $Q_{(m,\beta)}$ through order $M - d$. By (6.3), after multiplying by n^{-d} the error is bounded by n^{-M-1} times a fixed polynomial in m times $(\theta U)^m/m!$, with a constant depending only on the fixed parameters and β . The sum over m is finite uniformly in n . This is why no extra power of $\log n$ appears in the remainder.

We may now extend the retained dimer sums to $m = \infty$. Indeed, for each retained coefficient, (6.2) bounds the omitted summands by a polynomial in m times $(\theta U)^m/m!$. Since $k = m + K > T_n$ implies $m > T_n - K$, their tails are again $n^{-(2M+4)+o(1)}$. Summing each polynomial in m by (5.10) gives exactly (5.11). The total error is $O(n^{-M-1})$, uniformly on $|u| \leq U$. This proves (1.7) and, on setting $u = -1$, (1.11). $\square$

The proof also works for any fixed numbers r, s of paths whose shortest length is bounded below by a positive multiple of n . It does not assert uniformity for arbitrarily growing offsets or for forests with a bounded short path. [Added 1 October 2026, batch 74: Part II proves the expansion uniformly over forest shapes, on the disk $|1 + u| \leq 1$, whenever the shortest path has at least $K_n \asymp \log n / \log \log n$ vertices (theorem 16.6 and corollary 16.7); remark 16.8 notes that the proof above already gives this on every disk $|u| \leq U$ with T_n in place of K_n .]

7 Arithmetic, first corrections, and the full distribution

7.1 Degree and denominator control

Proposition 7.1 (Arithmetic of the correction polynomials). *The structural assertions (1.9) hold.*

Proof. For a defect- d profile, $C \leq d$ and $K \leq 2d$. A term in (5.11) has u -degree at most $K + 2(J - d) \leq 2J$. The formula is symmetric in r, s because $Q_{\mathbf{b}}$ is. For $J = 0$ it gives $B_0 = 1$. When $J > 0$, any nonempty β contributes positive u -degree. For the empty profile the constant term is $P_{\emptyset, J}(0) = 0$, since the completely empty $\mathbf{b}$ has $Q_{\mathbf{b}} = 1$. Thus $B_J(0) = 0$.

By Lemma 5.1, every $\Delta^t P(0)$ is an integer. A summand's only possible denominator is $t!\beta!$. This divides $(t + C)!$, by the multinomial coefficient, and

$$t + C \leq 2(J - d) + d \leq 2J. \quad (7.1)$$

Hence $t!\beta!$ divides $(2J)!$. Multiplying the finite sum by $(2J)!$ therefore gives a polynomial over $\mathbb{Z}$. $\square$

This denominator bound is unconditional. It is deliberately weaker than claiming all the oriented coefficients themselves are integers; that stronger phenomenon is discussed in section 11.

7.2 The first two corrections

For the dimer-only profile,

$$P_{\emptyset, 1}(m) = -m^2 + (2 - r - s)m = -m^2 + (1 - r - s)m. \quad (7.2)$$

The only defect-one profile is a single component of length three, with $C = 1$, $K = 2$, and $P_{\beta, 0} = 1$. Applying (5.10) immediately gives (1.10). For $\theta = 1$, a calculation of the four profile types of defect at most two gives

$$B_2^{(1)}(u; r, s) = (r - 1)(s - 1)u + \binom{r + s - 1}{2}u^2. \quad (7.3)$$

For completeness, the required dimer-only polynomial is

$$P_{\emptyset,2}(m) = \frac{1}{2}m^4 + (r+s-3)m^3 + \frac{r^2+2rs+s^2-5r-5s+7}{2}m^2 + \frac{-r^2+r-s^2+s}{2}m. \quad (7.4)$$

For one length-three component, $P_{\beta,1}(m) = -m^2 - (r+s)m + 3 - 2r - 2s$. The defect-two profiles are one length-four component or two length-three components; both have $P_{\beta,0} = 1$. These data and (5.11) verify (7.3) directly.

In the absolute case a useful general expression is

$$B_2^{(2)}(u; r, s) = 2(r-1)(s-1)u + 2((r+s)^2 - r - s - 1)u^2 + 2(2r+2s-1)u^3 + 2u^4. \quad (7.5)$$

Thus (1.5) is recovered as a special case of the first correction, while the theorem supplies every later order.

Remark 7.2 (Offsets (1,1): two classical sequences). [Added 1 October 2026, batch 73O2.] The manuscript does not discuss the smallest offsets, which give two classical sequences. The intake ran the shipped coefficient generator `code/path_forests.py` on a copy with `-r 1 -s 1 -order 10`.

- *Oriented case, no successions.* Here $A_{1,1}^{(1)}(n)$ counts permutations of $[n]$ with no $\pi(i+1) = \pi(i) + 1$, which is $A000255(n-1) = D_n + D_{n-1}$ with D_n the derangement numbers [10]. The formula gives $c_J^{(1)}(1,1) = 1, 1, 0, 0, \dots, 0$ for $J = 0, \dots, 10$. This is consistent with $(D_n + D_{n-1})/n! = e^{-1}(1 + 1/n) + O(1/n!)$, so every later coefficient is zero as well.
- *Absolute case, Hertzprung's problem.* Here $A_{1,1}^{(2)}(n)$ is A002464, the number of permutations with no rising or falling succession [11]. The formula gives

$$c_J^{(2)}(1,1) = 1, 0, -2, -\frac{10}{3}, -6, -\frac{154}{15}, -\frac{88}{9}, \frac{5336}{105}, \frac{1612}{3}, \frac{2098234}{567}, \frac{36500686}{1575} \quad (J = 0, \dots, 10).$$

These are, term for term, the Abramson–Moser expansion displayed in the OEIS entry (computed by Kotesovec, 2011, extended 2020), through n^{-10} .

Both cases agree with $c_1 = \theta(r+s-\theta)$ of (1.11). As an independent check, the intake computed both sequences exactly to $n = 800$, from the recurrences printed in the two OEIS entries (themselves checked against brute-force enumeration for $n \leq 8$). It found the scaled residuals $n^{M+1}(e^\theta A(n)/n! - \sum_{j \leq M} c_j n^{-j})$ for $M = 4, 8$ approaching c_{M+1} at $n = 200, 400, 800$. These finite checks are evidence; the proof is [theorem 1.1](#). The theorem covers the pair (1,1), and the manuscript's brute-force tests already include it (it is one of the six offset pairs of [section 10](#)). Only the comparison with the two OEIS entries is new here.

7.3 Poisson corrections for all probabilities

The zero-violation event is only one coefficient of the probability generating function. Write $B_J(u) = \sum_h b_{J,h} u^h$ and define

$$H_h(q; \theta) = \sum_{i=0}^{\min(h,q)} \binom{h}{i} (-1)^{h-i} \frac{q^i}{\theta^i}. \quad (7.6)$$

These are a normalization of the Charlier correction polynomials. Direct coefficient extraction gives

$$[t^q](t-1)^h e^{\theta(t-1)} = e^{-\theta} \frac{\theta^q}{q!} H_h(q; \theta). \quad (7.7)$$

Consequently define the signed approximation

$$p_M(q) = e^{-\theta} \frac{\theta^q}{q!} \sum_{J=0}^M \frac{1}{n^J} \sum_h b_{J,h} H_h(q; \theta). \quad (7.8)$$

Corollary 7.3 (An all-orders approximation of the entire law). *For fixed M, r, s, θ ,*

$$\sum_{q \geq 0} \left| \mathbb{P}(X_{r,s,n}^{(\theta)} = q) - p_M(q) \right| = O(n^{-M-1}). \quad (7.9)$$

In particular the law converges in total variation to $\text{Poisson}(\theta)$, with error $O(n^{-1})$.

Proof. Apply [theorem 1.1](#) on $|t| = R > 1$, using $u = t - 1$. Cauchy's inequality gives an error $O(n^{-M-1}R^{-q})$ in the coefficient of t^q . Summing the geometric series proves (7.9). $\square$

For $M \geq 1$, p_M need not be nonnegative at every q , so it is a signed asymptotic approximation, not a new probability model. It nevertheless has total mass one, because $B_0(0) = 1$ and $B_J(0) = 0$ for $J > 0$.

Remark 7.4 (A clique-model analogue). [Added 1 October 2026, batch 73O2, reciprocal note.] The companion report [a330266-balanced-smirnov-poisson](#) of this collection (batch 73, manuscript 57) studies the number $X_{n,k}$ of equal-rank adjacencies in a uniform shuffle of kn cards, k in each of n ranks. There the value relation is n cliques K_k , not a fixed number of long paths, so the hypotheses of this report do not apply, and the limit is $\text{Poisson}(\lambda)$ with $\lambda = k - 1$. Its first-order local law (Corollary 8.3 there, label `bsw:cor:local`),

$$\mathbb{P}(X_{n,k} = j) = e^{-\lambda} \frac{\lambda^j}{j!} \left[1 - \frac{(\lambda - j)^2 - j}{2kn} + O_{k,j}(n^{-2}) \right],$$

is the analogue of the case $M = 1$ of (7.8)–(7.9). That report proves it at first order only, and the analytic step behind its expansion is sketched; its Remark 6.3 names a uniform factorial-moment bound of the type of Lemma 4.1 as the device that would make it rigorous.

8 The two OEIS expansions, unconditionally

8.1 A189281: oriented distance-two avoidance

The first correction polynomials for $r = s = 2$, $\theta = 1$ are

$$B_0 = 1, \quad B_1 = -3u, \quad B_2 = u + 3u^2, \quad (8.1)$$

$$B_3 = u + u^2 - u^3, \quad B_4 = u - u^2 - 2u^3, \quad B_5 = u + u^2 - 3u^3, \quad (8.2)$$

$$B_6 = u + 23u^2 - 4u^3, \quad B_7 = u + 121u^2 + 19u^3. \quad (8.3)$$

Further examples are

$$B_8 = u + 479u^2 + 378u^3 + 24u^4, \quad (8.4)$$

$$B_9 = u + 1681u^2 + 3737u^3 + 672u^4 + 24u^5, \quad (8.5)$$

$$B_{10} = u + 5543u^2 + 28936u^3 + 11088u^4 + 984u^5 + 24u^6. \quad (8.6)$$

Setting $u = -1$ in these polynomials proves

$$\begin{aligned} \frac{\text{A189281}(n)}{n!} &= e^{-1} \left(1 + \frac{3}{n} + \frac{2}{n^2} + \frac{1}{n^3} + \frac{3}{n^5} + \frac{26}{n^6} + \frac{101}{n^7} + \frac{124}{n^8} \right. \\ &\quad \left. - \frac{1409}{n^9} - \frac{13266}{n^{10}} + O(n^{-11}) \right). \end{aligned} \quad (8.7)$$

The coefficient of n^{-4} is exactly zero. The terms through n^{-10} agree with the recurrence-dependent list in the OEIS [1]; the proof above does not assume that recurrence.

8.2 A110128: absolute distance-two avoidance

For $r = s = 2$, $\theta = 2$, the first polynomials are

$$B_1 = -6u - 2u^2, \quad (8.8)$$

$$B_2 = 2u + 22u^2 + 14u^3 + 2u^4, \quad (8.9)$$

$$B_3 = 2u + 6u^2 - 60u^3 - 56u^4 - 16u^5 - \frac{4}{3}u^6, \quad (8.10)$$

$$B_4 = 2u - 18u^2 - 122u^3 + 74u^4 + 154u^5 + 70u^6 + 12u^7 + \frac{2}{3}u^8. \quad (8.11)$$

It follows that

$$\begin{aligned} \frac{\text{A110128}(n)}{n!} = e^{-2} & \left(1 + \frac{4}{n} + \frac{8}{n^2} + \frac{68}{3n^3} + \frac{242}{3n^4} + \frac{1692}{5n^5} \right. \\ & + \frac{72802}{45n^6} + \frac{2725708}{315n^7} + \frac{16083826}{315n^8} \\ & \left. + \frac{186091480}{567n^9} + \frac{32213578294}{14175n^{10}} + O(n^{-11}) \right). \end{aligned} \quad (8.12)$$

This recovers the higher list attributed to the guessed operator in the OEIS [2], again by a direct argument from the permutation model. The fractions here also show why an integer-coefficient assertion cannot be transferred indiscriminately from the oriented case to the absolute case.

8.3 Additional coefficients and how to use them

J	$c_J^{(1)}(2, 2)$	$c_J^{(2)}(2, 2)$
11	-59103	2620164054016/155925
12	-9448	61699558213516/467775
13	2525459	6658719787519636/6081075
14	26632378	136086447719175784/14189175
15	152270749	5133067332126139204/58046625
16	106221948	546416517735316663856/638512875

Table 1: Exact evaluations of the finite coefficient formula beyond the order-ten lists. The accompanying JSON includes every polynomial B_J through order sixteen, not only its value at $u = -1$.

For any fixed M , adding terms through $c_M n^{-M}$ gives a remainder $O(n^{-M-1})$ inside the parentheses. This is a Poincaré statement with M fixed as $n \rightarrow \infty$. It does not establish convergence of the infinite series, justify setting M proportional to n , or identify an optimal truncation rule. The exact finite coefficients in [table 1](#) are supplied with their generating code rather than inferred from a numerical fit.

9 Lambert- W inversion with a discrete interpretation

Remark 9.1 (Relation to the repository’s inversion apparatus). [Added 1 October 2026, batch 7302.] This section re-derives, for its own forward map, machinery that the canonical volume *Transseries and Inversion* of the ProveIt repository already contains [12], and the manuscript does not cite it. The forward map $e^{-\theta} \Gamma(n+1) C(1/n)$ is the gamma carrier of that volume’s chapter on the increasing inverse of Gamma (theorem “All-orders expansion of Γ_+^{-1} ”, label `p6:thm:gamma`),

perturbed by a power series in $1/n$, which is the situation of its theorem “Perturbed inversion” (`p0:thm:perturbed-inversion`). The volume’s chapter “The subfactorial” (`p8:sec:top`) treats the closest analogue, the derangement numbers, by citing that carrier. Normalizations differ: here $z = L/W_0(L/e)$ with $L = \log y$, and the constant $\frac{1}{2} \log(2\pi)$ sits in κ_θ ; the volume uses $R = \log y - \frac{1}{2} \log(2\pi)$ and $T = R/W_0(R/e)$. The discussion of integer thresholds at the end of the section is an instance of the volume’s staircase theorem (`p0:thm:staircase`). [Theorems 9.2](#) and [9.3](#) are therefore printed as a self-contained second route for this family of sequences. No novelty is claimed for the inversion method; what is specific to this report is the forward input, the coefficients c_j of [theorem 1.1](#).

9.1 Logarithmic coefficients

For either sequence, or any fixed r, s, θ , suppress the parameters and write

$$\frac{A(n)}{n!} \sim e^{-\theta} C(1/n), \quad C(x) = 1 + \sum_{j \geq 1} c_j x^j. \quad (9.1)$$

The right side is a formal asymptotic series. Define $\log C(x) = \sum_{j \geq 1} \ell_j x^j$. An explicit coefficient recursion is

$$\ell_j = c_j - \frac{1}{j} \sum_{k=1}^{j-1} k \ell_k c_{j-k}. \quad (9.2)$$

Combining the proved probability expansion with Stirling’s expansion, in its usual remainder-controlled form [7], yields

$$\log A(n) = n \log n - n + \frac{1}{2} \log n + \kappa_\theta + \sum_{j=1}^M a_j n^{-j} + O(n^{-M-1}), \quad (9.3)$$

where

$$\kappa_\theta = \frac{1}{2} \log(2\pi) - \theta, \quad a_j = \ell_j + \frac{B_{j+1}}{j(j+1)}. \quad (9.4)$$

Here B_m are Bernoulli numbers; the additional term vanishes for even j . For A189281,

$$\ell_1 = 3, \quad \ell_2 = -\frac{5}{2}, \quad \ell_3 = 4, \quad \ell_4 = -\frac{29}{4}, \quad a_1 = \frac{37}{12}, \quad a_2 = -\frac{5}{2}, \quad a_3 = \frac{1439}{360}. \quad (9.5)$$

For A110128, $a_1 = 49/12$.

9.2 The first explicit inverse terms

Given $y = A(n)$, let $L = \log y$ and

$$z = \frac{L}{W_0(L/e)}, \quad w = \log z. \quad (9.6)$$

The principal Lambert function is characterized by $W_0(x)e^{W_0(x)} = x$ for $x > 0$ [8]. Thus $z(\log z - 1) = L$ exactly.

Theorem 9.2 (First inverse corrections). *Along the sequence values $y = A(n)$ as $n \rightarrow \infty$,*

$$n = z - \frac{1}{2} - \frac{\kappa_\theta}{w} - \frac{1}{z} \left(\frac{c_1 - 1/24}{w} + \frac{\kappa_\theta^2}{2w^3} \right) + O\left(\frac{1}{z^2 w}\right). \quad (9.7)$$

For A189281 the coefficient $c_1 - 1/24$ is $71/24$; for A110128 it is $95/24$.

Proof. Set $n = z + D$. The derivative of $x \log x - x$ at z is w , and its second derivative is $1/z$. The constant-order terms of (9.3), after subtracting L , give $wD + \frac{1}{2}w + \kappa_\theta = 0$, hence $D = d_0 = -1/2 - \kappa_\theta/w$ at that order. The coefficient of z^{-1} is

$$wd_1 + \frac{d_0^2 + d_0}{2} + a_1. \quad (9.8)$$

Since $a_1 = c_1 + 1/12$, this gives

$$d_1 = -\frac{c_1 - 1/24}{w} - \frac{\kappa_\theta^2}{2w^3}. \quad (9.9)$$

The residual in the logarithmic equation is $O(z^{-2})$; division by a derivative asymptotic to w gives the asserted index error. A precise all-orders version of this argument follows next. $\square$

9.3 An all-orders inverse recursion

Put $t = 1/z$ and seek $D(t, w) = \sum_{j \geq 0} d_j(w)t^j$. Define

$$\begin{aligned} \mathcal{H}(t, D) = & \frac{(1 + tD) \log(1 + tD) - tD}{t} + \frac{1}{2} \log(1 + tD) \\ & + \sum_{j \geq 1} a_j t^j (1 + tD)^{-j}. \end{aligned} \quad (9.10)$$

The quotient by t is a formal series divisible by t , so no singular constant term occurs. The inverse equation is

$$wD + \frac{1}{2}w + \kappa_\theta + \mathcal{H}(t, D) = 0. \quad (9.11)$$

It gives the triangular recursion

$$d_0(w) = -\frac{1}{2} - \kappa_\theta/w, \quad (9.12)$$

$$d_j(w) = -\frac{1}{w} [t^j] \mathcal{H} \left(t, \sum_{q=0}^{j-1} d_q(w)t^q \right) \quad (j \geq 1). \quad (9.13)$$

Only $a_1, \dots, a_j$ enter the j th step.

Theorem 9.3 (All-orders index inversion). *For every fixed $J \geq 0$, the functions in (9.13) are rational in w , with coefficients in $\mathbb{Q}[\kappa_\theta]$ for the rational forward coefficients considered here, and*

$$n = z + \sum_{j=0}^J \frac{d_j(w)}{z^j} + O\left(\frac{1}{z^{J+1}w}\right), \quad y = A(n). \quad (9.14)$$

Moreover, $d_0 = O(1)$ and $d_j = O(1/w)$ for $j \geq 1$ as $w \rightarrow \infty$.

Proof. The recursion first proves the algebraic form of the coefficients by induction. Since $\mathcal{H}$ has no t^0 term, its coefficient of t^j depends only on the earlier d_q , and stays bounded when those earlier coefficients are bounded. Dividing by w proves the stated size bounds.

Fix J and retain sufficiently many terms of (9.3). Substitution of the truncated D cancels every term through z^{-J} , leaving a residual $O(z^{-J-1})$. All relevant derivatives of the finite logarithmic model are controlled on a bounded neighborhood of $z + d_0$, and its first derivative is $\log z + O(1/z)$. The actual index is in such a neighborhood: the leading term first gives $n/z \rightarrow 1$, and then the mean value theorem applied to $n \log n - n - z \log z + z = O(\log n)$ gives $n - z = O(1)$. A further application of the mean value theorem divides the residual by a quantity asymptotic to w , proving (9.14). $\square$

9.4 What is being inverted

A discrete sequence has no distinguished analytic interpolation. The preceding theorems make no such choice: their statements are evaluated at the actual values $y = A(n)$. Alternatively, define an explicit smooth approximation

$$\mathcal{A}_M(x) = e^{-\theta} \Gamma(x+1) \sum_{j=0}^M c_j x^{-j}. \quad (9.15)$$

It is positive and strictly increasing for all sufficiently large x , because the finite correction tends to one and the logarithmic derivative is $\log x + O(1/x)$. Its inverse satisfies

$$\mathcal{A}_M^{-1}(A(n)) - n = O\left(\frac{n^{-M-1}}{\log n}\right). \quad (9.16)$$

This is a well-defined model inverse, with its discrepancy from the sequence controlled.

For an integer threshold $N(y) = \min\{n : A(n) \geq y\}$, an asymptotic estimate alone does not justify unconditional nearest-integer rounding near a jump. Certified bounds for $A(n)$ can resolve the remaining adjacent candidates. The Bonferroni method in the next section provides one such tool. [Added 1 October 2026, batch 73O2: the general form of this remark, exact rounding $N(y) = \lceil \nu(y) \rceil$ for an increasing interpolation ν and the separation condition under which an approximate inverse decides it, is the staircase theorem `p0:thm:staircase` of [12]. Its separation step is formalized in Lean as `Fabius.staircase_separation` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`. That formalization is context only; it does not verify any statement of this report.]

10 Reproducible computations and exact certificates

10.1 Coefficient generation

The core implementation, `code/path_forests.py`, uses only the Python standard library. It forms the numerator factors and reciprocal denominator in (5.3) as truncated integer polynomials; evaluates $P_{\beta,j}(m)$ at the proved sufficient set of integers; takes forward differences; and assembles (5.11) using exact rational arithmetic. The work depends on the requested asymptotic order, not on a large sequence index n .

At order J , profiles are indexed by partitions of $d \leq J$, and each profile uses at most $2(J-d)+1$ dimer evaluations. No integer-sequence terms are used as fitting data. The archive contains all B_J and c_J through order sixteen in `data/coefficients_order16.json`.

10.2 Independent finite checks

The included validation program completed the following checks.

Check	Exact equalities/cases
Full tiling enumeration against both OEIS lists, $n = 0, \dots, 21$	44
Brute-force full distributions for $n = 1, \dots, 8$, six offset pairs, and both statistics	96
Stable factorial moments against those brute-force distributions	246
Stable tiling polynomial on specified unequal path-length forests	86
Additional finite-difference tests beyond the interpolation degree	375
Factorial denominator checks on polynomial coefficients through order 16	396

The six offset pairs were $(1, 1), (1, 2), (2, 2), (2, 3), (3, 2), (3, 3)$. Full tiling enumeration is independent of the stable-polynomial implementation, and brute permutation enumeration is independent of both. These checks detect finite implementation and bookkeeping errors; none replaces the proofs for arbitrary n or arbitrary expansion order.

10.3 Bonferroni probability intervals

When (3.13) holds through $2q + 1$, exact factorial moments give

$$S_{2q+1} \leq \mathbb{P}(X = 0) \leq S_{2q}, \quad S_j = \sum_{k=0}^j (-1)^k \mu_k. \quad (10.1)$$

For completeness, this follows pointwise from $\sum_{k=0}^j (-1)^k \binom{x}{k} = (-1)^j \binom{x-1}{j}$ when $x \geq 1$, while both partial sums equal one for $x = 0$. Thus the lower and upper bounds are rigorous rational numbers, even though the full count has not been computed.

The certificate program uses S_{31} and S_{30} at $n = 80, 160, 320$. It brackets e^{-1} between the odd and even Taylor sums of degrees 101 and 102, and squares that interval for e^{-2} . All arithmetic in the resulting error intervals is rational. The quantity displayed below is

$$E_M(n) = n^{M+1} \left(e^{\theta} \frac{A_{2,2}^{(\theta)}(n)}{n!} - \sum_{j=0}^M c_j n^{-j} \right). \quad (10.2)$$

By the theorem its limit is c_{M+1} .

θ	n	Certified interval for $E_4(n)$	Certified interval for $E_8(n)$
1	80	[3.3409847, 3.3409848]	[-1584.0080, -1584.0079]
1	160	[3.1664733, 3.1664734]	[-1494.2195, -1494.2194]
1	320	[3.0822399, 3.0822400]	[-1451.0335, -1451.0334]
2	80	[360.08331, 360.08332]	[359523.99, 359524.00]
2	160	[348.86238, 348.86239]	[343097.56, 343097.57]
2	320	[343.54178, 343.54179]	[335473.66, 335473.68]

Table 2: Outward-rounded exact intervals. The limits are $3, -1409$ for $\theta = 1$, and $1692/5, 186091480/567$ for $\theta = 2$. The intervals certify these finite residuals; their convergence is a consequence of the proof, not an inference from three samples.

The archive includes the unrounded rational endpoints and additional order-ten residual certificates. With the same moment cutoff, those order-ten intervals become wider at larger n , especially for $\theta = 2$: the factor n^{11} magnifies the fixed Bonferroni width. This is an honest precision limit of that finite computation; increasing the cutoff addresses it without using a guessed recurrence.

10.4 Inverse diagnostics and reproducibility

Using exact enumerated sequence values at $n = 20$, the Lambert backbone z has index errors about 0.52023 for A189281 and 0.20749 for A110128. Including both displayed correction levels in (9.7) reduces these to approximately -0.00066084 and 0.00089419 , respectively. These are numerical diagnostics, not certified rounding guarantees.

The programs were run with Python 3.13.5. The core coefficient, enumeration, and rational-certificate programs need no third-party package. The optional rational-identity exploration uses SymPy 1.14.0; the inverse diagnostic uses mpmath 1.3.0. The commands and expected

outputs are recorded in the archive's README. No Lean code was executed or supplied as an asserted formal proof. [Added 1 October 2026, batch 73O2: the delivered README is not shipped. The report's README.md replaces it and gives rerun commands for the shipped layout; see [section B](#).]

11 A sharper conjecture suggested by the exact computations

There is a striking simplification in the oriented $(2, 2)$ case that is not needed by any preceding theorem. Define the universal rational moments

$$\mu_k^*(n) = \sum_{\mathbf{b}: \sum (\ell-1)b_\ell = k} \frac{F_2(n; \mathbf{b})^2}{\mathbf{b}! n^{\underline{v}}}, \quad (11.1)$$

as rational functions in an indeterminate n . They agree with the actual moments whenever the stability conditions hold. Remove the leading Poisson factor by setting

$$R_h(n) = \sum_{k=0}^h \frac{(-1)^{h-k}}{(h-k)!} \mu_k^*(n). \quad (11.2)$$

Exact finite algebra gives

$$R_0 = 1, \quad (11.3)$$

$$R_1 = -\frac{3n-4}{n^2}, \quad (11.4)$$

$$R_2 = \frac{3n^2 - 17n + 26}{n^4}, \quad (11.5)$$

$$R_3 = -\frac{(n-6)(n^2 - 7n + 16)}{n^6}. \quad (11.6)$$

Conjecture 11.1 (Rational collapse). For every $h \geq 4$,

$$R_h(n) = \frac{24(n-h+1)^{h-4}}{n^{2h}} = \frac{24}{n^{h-1}(n-2h+5)^5}. \quad (11.7)$$

This has been checked as an exact identity of rational functions for each $h = 4, \dots, 16$. The check clears the common denominator n^{2h} and compares polynomials; it is stronger than testing finitely many numerical values of n , but it is still only a finite check in h . No induction or creative-telecoping certificate for all h has been obtained here.

The proposed formula would imply the recurrence

$$\frac{R_{h+1}(n)}{R_h(n)} = \frac{(n-2h+5)(n-2h+4)}{(n-h+1)(n-2h)(n-2h-1)} \quad (h \geq 4). \quad (11.8)$$

Proving (11.8) directly from (11.2) is therefore a sharply specified next target.

Proposition 11.2 (What the rational collapse would imply). *If Conjecture 11.1 holds, then $B_J^{(1)}(u; 2, 2) \in \mathbb{Z}[u]$ for every J , and therefore all $c_J^{(1)}(2, 2)$ are integers. For $J \geq 4$ it would also give*

$$\deg B_J^{(1)}(u; 2, 2) \leq \max(3, J-4). \quad (11.9)$$

Proof. The first four rational functions $R_h(1/x)$ have integer Taylor coefficients. Under the conjecture, each later one is $24x^{h+4}$ times a product of factors $(1-ax)^{-1}$ with nonnegative integers a , after cancellation. It too has integer coefficients and begins at degree $h+4$. At any fixed power x^J , only finitely many of these R_h can contribute.

By (11.2), the coefficient of u^h in the Poisson-normalized stable moment series is $R_h(n)$. Comparing with the already proved coefficient formula gives $B_J^{(1)}(u; 2, 2) = \sum_h [x^J] R_h(1/x) u^h$. The integrality and degree claims follow. $\square$

This conditional proposition must not be confused with the unconditional factorial denominator theorem. The conjecture packages the observed integrality into a more structured rational-function identity, making it more approachable than a bare statement about an infinite list of integers.

12 Further research questions and proposed next steps

1. Prove or refute the rational collapse. The most concrete next step is an identity in two symbolic parameters rather than a fit to sequence values. Clear denominators in (11.2), organize profiles by their component count, and seek a telescoping proof of (11.8). A combinatorial cancellation proof would be particularly valuable: it should explain both the factor 24 and the loss of four powers of n beyond the moment index. A failure at some later h would also be informative, because the first discrepancy would identify a missing component interaction. The all-orders integrality issue for the oriented family remains separate until such a proof exists. [Added 1 October 2026, batch 74: for the oriented one-path subfamily, $r = 1$ or $s = 1$, all-orders integrality is proved in Part II (corollary 18.2); the case $(2, 2)$ of A189281 remains open.]

2. Connect the asymptotic theorem to the specific guessed operators. A proved asymptotic expansion does not establish the A189281 or A110128 recurrence. Conversely, a rigorous operator certificate could supply a second, independent route to large-order coefficient behavior. One useful project is to translate the stable moment framework into a finite-dimensional holonomic representation with boundary terms retained, then compare its annihilator with the listed operators. All boundary terms must be proved, not discarded because their numerical influence is small.

3. Identify the first geometry-dependent exponential sector. Theorem 4.2 shows that long path-length differences are invisible to every algebraic order. It does not give their leading amplitude, sign, or possible dependence on n modulo the offsets. The first profiles that can see a short boundary have edge degree comparable to the shortest path. A saddle-point analysis of that moving profile boundary could reveal the first exponentially small sector, and would turn the present algebraic expansion into part of a genuine transseries description.

4. Determine large-order growth and justified optimal truncation. The coefficients through order sixteen already show that sign and magnitude can change markedly. The present proof allows constants to depend on M . It would be useful to prove explicit growth bounds in M , decide whether a Gevrey-type estimate is sharp, and derive a valid truncation rule for a given n . The rational-collapse conjecture, if proved, would make the oriented case a tractable testing ground, but the absolute case needs its own analysis.

5. Allow offsets or path counts to grow with n . The current proof uses fixed r, s , both in the shortest-path condition and in the analytic bound on $Q_{\mathbf{b}}$. When r or s grows, the shortest path may meet the moment cutoff, and the number of path boundaries changes the first corrections. Determine the largest growth range in which a useful uniform expansion persists, and the transition law when path lengths become comparable to $\log n / \log \log n$. This question requires new uniform estimates; it is not an immediate corollary of the fixed-offset theorem. [Added 1 October 2026, batch 74: for fixed r, s , Part II shows that the expansion survives whenever the shortest path is long compared with $\log n / \log \log n$ (corollary 16.7); the transition at that scale remains open (section 22, question 2).]

6. Extend from paths to several forbidden offsets and to cycles. For several allowed differences, selected components can branch or contain cycles. Their embeddings are no longer determined by one starting value and a single orientation. A useful extension would classify component types by an appropriate defect and prove an analogue of the uniform moment bound and finite coefficient formula. Cyclic boundaries are another natural test: short cycles can contribute at a finite algebraic order, whereas long cycles may only change a beyond-all-orders sector.

7. Formalize the proof and build a certified coefficient library. The finite profile identity, the stable tiling polynomial, and the integer coefficient arithmetic are natural first formalization targets. The analytic layer then needs the uniform factorial-moment tail, Cauchy bounds, and finite asymptotic truncation. A robust implementation should expose the exact stability hypotheses instead of silently applying F_r outside its range. The rational Bonferroni certificates are suitable artifacts for an independent checker. This is a proposed formalization route, not a report of completed Lean verification.

8. Combine asymptotic inversion with exact threshold certification. For very large y , the Lambert backbone and a few correction terms should reduce the inverse problem to a small number of candidate indices. Use exact Bonferroni intervals to compare each candidate count with y , adaptively increasing the moment cutoff only when necessary. The interesting algorithmic question is a proved cost bound that accounts for near-threshold inputs, where asymptotic rounding alone cannot decide the answer.

13 Conclusion

The main result is a direct all-orders theorem for a natural pair of permutation sequences whose higher displayed OEIS expansions were tied to guessed recurrences. Its proof does not solve those recurrence problems by assumption: it bypasses them. The stable path-tiling identity gives exact low moments, the factorial tail controls the remaining moments, and finite differences sum the unlimited dimer contributions. Together they turn a formal asymptotic calculation into a uniform theorem.

The same structure yields more than two lists of numbers: a full-law Poisson correction expansion, a quantitative explanation of boundary-length universality, an arithmetic denominator theorem, and a rigorous inverse construction. The stronger rational-collapse pattern is separated from these proved results and posed as a specific next problem. That distinction is central to the usefulness of the report: the established theorems can be used now, while the remaining conjectures have explicit, testable targets.

Part II

Fixed-gap asymptotics: a second derivation and its own results

14 Provenance, scope and notation of Part II

[Added 1 October 2026, batch 74.] Part II was added in the write of batch 74. It is built from manuscript 03 of that batch and prints only what that manuscript adds to Part I. Where the two manuscripts prove the same statement, the statement is printed once, in Part I, and

manuscript 03’s route is recorded in [section 15](#). Everything in Part II is written in Part I’s notation ([table 4](#)).

14.1 Provenance

Manuscript 03 of batch 74 is the archive `ProveIt_Fixed_Gap_Asymptotics.zip`, which arrived in commit `c664fc2f0` and was placed in this report by commit `b669cff87`. Its article is titled *All-Orders Asymptotics and Boundary Universality for Fixed-Gap Permutations* (25 pages, dated October 1, 2026); its title page reads “Prepared for Vladimir Reshetnikov” and its PDF metadata names “ChatGPT, research report prepared for Vladimir Reshetnikov” as author. Its pinned repository commit is `291fbe6bb4477e0cb2131f1e041b12a9abe363fd` (2026-10-01, 18:10; [\[14\]](#)).

It was written without sight of manuscript 60, the source of Part I, and proves Part I’s main theorems in independent text: its structure, notation, constants and several proof devices differ throughout, and the two texts share about 1.1 % of their word 8-grams, almost all of them bibliography entries and the digits of the coefficient list. Every number the two manuscripts share agrees exactly: the coefficients of A189281 and A110128 through order twelve, the general second correction in r, s for both orientations, the polynomials B_1, B_2, B_3 at $(r, s) = (2, 2)$, the logarithmic coefficients and the first inverse coefficients.

[Added 1 October 2026, batch 74: manuscript 03 states that “a repository code search for A189281 returned no match at the time of the audit”, and its delivered `02-fixed-gap-sources.md` repeats it. The pinned commit `291fbe6bb` already held the archive of manuscript 60, `docs/incoming/oeis_path_forests.zip` (arrival commit `aa43cc555`), which code search cannot read; that archive was placed an hour later as Part I of this report (commit `6e193dd4f`). Part II is an independent derivation of the same theorems plus the results listed below.]

Scope and non-claims of manuscript 03, kept. Its contribution is “the explicit stable-profile and finite-defect framework, the uniform error analysis, and the quantitative comparison of whole laws”. The leading Poisson behaviour, Tauraso’s first correction and the matching-of-tilings inclusion–exclusion of Spahn and Zeilberger are credited as prior work. The proofs are ordinary mathematics, not a Lean-certified development, and no publication priority is claimed. “All orders” means a remainder theorem at each fixed order, not a convergent or resurgent completion; no Borel summability, optimal truncation, Stokes constant or complete family of exponentially small sectors is asserted. The conjectured OEIS recurrences of A189281 and A110128 are not proved, and neither is all-orders integrality when both component counts exceed one. Its numerical tables are not interval certificates, and its values of A189281 at $n = 80, 120$ are b-file inputs, not regenerated. It made no OEIS submission and modified no repository file.

14.2 What Part II adds, and how it compares with Part I

14.3 Notation

Manuscript 03 and Part I use several letters for different objects, and manuscript 03 overloads D, w, T, A and B internally. [Table 4](#) fixes the convention: Part II uses Part I’s names for every shared object and new names $(K_n, D_{n,k}, \Lambda, N, \Delta, \omega_j, \varepsilon_M, \alpha)$ for the rest. No normalization is changed.

Topic	Part I (manuscript 60)	Part II (manuscript 03)
All-orders remainder	fixed offsets r, s ; uniform on every disk $ u \leq U$ (theorem 1.1)	only on $ 1 + u \leq 1$, i.e. $ u \leq 2$, but uniform over all forest shapes whose shortest path has at least $K_n \asymp \log n / \log \log n$ vertices (theorem 16.6 and corollary 16.7)
Coefficients	B_J through order 16, $(2J)!B_J \in \mathbb{Z}[u]$ (Proposition 7.1)	same values through order 12 (not new); B_J polynomial in r, s of total degree at most J (proposition 17.1); closed forms of $c_3^{(\theta)}(r, s)$ (proposition 17.2)
Integrality	for $(2, 2)$ oriented only conditional on Conjecture 11.1	<i>proved</i> for the oriented case with $r = 1$ or $s = 1$ (corollary 18.2); $(2, 2)$ untouched
Universality	total variation bound (theorem 4.2)	bounds for the whole generating function and the avoidance probability, a second total-variation bound, and an $n = 12$ example (theorem 16.3 and remark 16.4)
Whole law	ℓ^1 approximation of the law (Corollary 7.3)	explicit first-order local laws at $(2, 2)$ (19.2)–(19.3)
Inverse	all orders in the coordinates z, w (theorem 9.3)	explicit terms through N^{-3} in a coordinate absorbing κ_θ ; a_2, a_3 for A110128 (proposition 20.1)
Certificates	certified Bonferroni intervals (table 2)	uncertified decimals only (section 21)

Table 3: Part II against Part I. The remainder theorem of Part II is weaker in u and stronger in geometry; see also [remark 16.8](#).

Notion	Part I and Part II	Manuscript 03
orientation weight (1 or 2)	θ	η (Part I's η is a length proportion)
length proportion	α (Part II); η in theorem 4.2	α
generating function $\mathbb{E}\binom{X}{k}$	$G(1+u) = \mathbb{E}(1+u)^X$ μ_k	$G(w)$, $w = 1+z$, $z = u$ S_k (Part I's S_j are Bonferroni sums)
profile, multiplicity factorial	$\mathbf{b} = (b_\ell)$, $\mathbf{b}!$	$\mathbf{m} = (m_j)$, $\mathbf{m}!$
tiling count	$C_{\mathcal{P}}(\mathbf{b})$	$T_{\mathcal{F}}(\mathbf{m})$
stable tiling polynomial	$F_r(n; \mathbf{b})$	$\mathbf{b}!^{-1} F_r$ is its $T_r(n; \mathbf{m})$
formal root	$1+W$, $\mathcal{D}^{-1}$	λ , A
dimers; long profile	m ; β , C , K	h ; $\mathbf{m}$, c_0 , k_0
normalized profile kernel	$Q_{\mathbf{b}}(x)$	$H_h(t)$ (<i>not</i> Part I's H_h)
Charlier polynomials	$H_h(q; \theta)$	$Q_m(j; \eta)$ (<i>not</i> Part I's $Q_{\mathbf{b}}$)
correction polynomials	$B_J^{(\theta)}(u; r, s)$	$C_J(r, s; \eta, z)$
avoidance coefficients	$c_J^{(\theta)}(r, s)$	$c_J^{(\eta)}(r, s)$ (same)
moment cutoff $(1 - 2k/n)^{-2k}$	T_n (Part I), K_n (Part II) $D_{n,k}$	K (Part I's K is an edge degree) $D_{n,k}$ (its D is also $\log N$)
logarithmic coefficients	a_j	d_j (Part I's d_j are inverse coefficients)
$\frac{1}{2} \log(2\pi) - \theta$	κ_θ	γ
inverse coordinates	Λ , N , $\Delta = \log N$ (Part II)	L , N , D
inverse coefficients	$\omega_j(\Delta)$	$u_j(D)$
unscaled truncation error	$\varepsilon_M(n)$	$E_M(n)$ (Part I's E_M is scaled by n^{M+1})

Table 4: Notation of manuscript 03 against Part I. Part II uses the middle column throughout.

15 Manuscript 03's second routes to Part I's theorems

Table 5 records, for each result of Part I that manuscript 03 proves again, the device its proof uses. None of these statements is reprinted. Where a device is also needed for a new result of Part II, it is stated there.

Part I	Manuscript 03's route
Proposition 2.2	the same matching-of-tilings argument, stated for binomial moments μ_k , with the same orientation lemma (weight θ^c , not θ^k); adds the symmetry $A_{r,s} = A_{s,r}$ via $\pi \mapsto \pi^{-1}$
(3.6)	a formal residue identity for the root $\lambda = 1 + \sum_{\ell} z_{\ell} \lambda^{1-\ell}$, with $A = (1 + \sum_{\ell} (\ell-1) z_{\ell} \lambda^{-\ell})^{-1}$; here $\lambda = 1 + W$ and $A = \mathcal{D}^{-1}$
theorem 3.1	the identical polynomial, written $T_r(n; \mathbf{m}) = F_r(n; \mathbf{b})/\mathbf{b}!$
Lemma 4.1	$\mu_k \leq \theta^k D_{n,k}/k!$ for $1 \leq k < n/2$ (lemma 16.1): sharper for $k \ll n$, Part I's holds for every k
(4.2)	a Taylor remainder on the unit disk $ 1+u \leq 1$ (lemma 16.2), not an exponential-series tail
theorem 4.2	total variation through Parseval and Cauchy–Schwarz (theorem 16.3); the $\exp(-\delta n \log n + O(n))$ form is Part I's last assertion with η' renamed δ
Definition 5.2	the same finite formula; the dimer sum is evaluated by Touchard polynomials, $e^{-x} \sum_m x^m m^q / m! = T_q(x)$ with $T_{q+1} = x(T_q + T'_q)$, instead of finite differences, and the denominator by exponentials of power sums
Proposition 7.1	$B_0 = 1$, $\deg_u B_J \leq 2J$, $B_J(0) = 0$ and symmetry; adds the degree in r, s (proposition 17.1); Part I's denominator theorem is not in manuscript 03
(1.10), (7.3), (7.5)	B_1 by the same two-profile hand calculation; general B_2 and B_3 in symbolic r, s in <code>data/02-fixed-gap-general_pgf_first_three.txt</code> ; B_2 agrees with Part I identically in r, s, u
theorem 1.1	theorem 16.6: a defect-sensitive moment bound and a kernel bound on $ x \leq 1/(D(1+k)^2)$ in place of Lemma 6.1
(8.7), (8.12), table 1	the coefficients of both sequences through order 12, equal to Part I's (see below)
residue classes (section 1)	the same explanation by theorem 3.1
(3.14)	the same count; adds $\mathbb{E}X = \theta + \theta(1-r-s)/n + \theta(r-1)(s-1) \sum_{q \geq 2} n^{-q}$
(9.3), theorems 9.2 and 9.3	the same logarithmic series; the inverse in a coordinate absorbing κ_{θ} (proposition 20.1)
section 9, last subsection	the same smooth carrier $\mathcal{A}_M$ and threshold bracket

Table 5: Results of Part I proved again by manuscript 03, and the device of each second route.

Order twelve is not new. Manuscript 03 presents its coefficients through order twelve as an extension of the OEIS lists (“extend them through order twelve”) and proposes appending

$$-\frac{59103}{n^{11}} - \frac{9448}{n^{12}} \text{ to A189281, } \frac{2620164054016}{155925 n^{11}} + \frac{61699558213516}{467775 n^{12}} \text{ to A110128.}$$

[Added 1 October 2026, batch 74: Part I's table 1 already gives orders 11 to 16 of both sequences. Manuscript 03's values at orders 0 to 12, computed independently by its own engine, agree with Part I's exactly; they confirm Part I's table and are not new.]

The inverse is not new as a method. As for Part I (Remark 9.1), manuscript 03's inverse is an instance of the transseries volume's apparatus [12]: its coordinates are those of `p6:thm:gamma` with $\log Y$ shifted by θ , its inverse expansion is a perturbed inversion around that gamma carrier (`p0:thm:perturbed-inversion`), and its threshold bracket is `p0:thm:staircase`. No novelty is claimed for the method.

16 Uniformity over forest shapes

Throughout this section $\mathcal{P}$ and $\mathcal{V}$ are arbitrary directed path forests on n vertices with r and s paths, as in section 2, and

$$D_{n,k} = \left(1 - \frac{2k}{n}\right)^{-2k} \quad (1 \leq k < n/2). \quad (16.1)$$

Lemma 16.1 (A sharper binomial-moment bound; manuscript 03). *For either statistic and arbitrary path forests, $\mu_k \leq \theta^k D_{n,k}/k!$ for $1 \leq k < n/2$.*

Proof (manuscript 03). As in Lemma 4.1, a selected set of k position edges with c components has probability at most $\theta^c n^c / n^{k+c}$. Since $c \leq k$ and $k + c \leq 2k$, this is at most $\theta^k n^{-k} D_{n,k}$; there are at most $n^k/k!$ such sets. $\square$

Lemma 16.2 (Taylor remainder on the unit disk; manuscript 03). *For $|1 + u| \leq 1$ and every integer $K \geq 1$,*

$$\left| G(1 + u) - \sum_{k=0}^{K-1} \mu_k u^k \right| \leq |u|^K \mu_K. \quad (16.2)$$

Proof (manuscript 03). For an integer $x \geq K$, Taylor's integral remainder for w^x about $w = 1$ along the segment from 1 to $w = 1 + u$, which stays in the closed unit disk, has modulus at most $\binom{x}{K} |u|^K$; for $x < K$ it vanishes. Take expectations. $\square$

Theorem 16.3 (Quantitative boundary universality; manuscript 03). *Let $(\mathcal{P}, \mathcal{V})$ and $(\tilde{\mathcal{P}}, \tilde{\mathcal{V}})$ be two pairs of forests on n vertices with r position and s value paths, every path of all four having at least K vertices, $1 \leq K < n/2$, and let $X, \tilde{X}$ be their statistics for the same θ . Then $\mu_k = \tilde{\mu}_k$ for $0 \leq k \leq K$, and*

$$\sup_{|w| \leq 1} |G(w) - \tilde{G}(w)| \leq \frac{2(2\theta)^K}{K!} D_{n,K}, \quad (16.3)$$

$$|\mathbb{P}(X = 0) - \mathbb{P}(\tilde{X} = 0)| \leq \frac{2\theta^K}{K!} D_{n,K}, \quad (16.4)$$

$$d_{\text{TV}}(\mathcal{L}(X), \mathcal{L}(\tilde{X})) \leq \sqrt{n+1} \frac{(2\theta)^K}{K!} D_{n,K}. \quad (16.5)$$

If every path has at least αn vertices, all three are at most $\exp\{-\delta n \log n + O_{\alpha, \delta, \theta}(n)\}$ for every fixed $0 < \delta < \min(\alpha, 1/2)$.

Proof (manuscript 03). Moments of order $k \leq K$ are stable in all four forests and agree by Corollary 3.2. Subtract the common Taylor polynomial of degree $K - 1$ and apply lemmas 16.1 and 16.2 to both remainders, with $|u| = |w - 1| \leq 2$ on the disk and $|u| = 1$ at $w = 0$. For the total variation, write $G - \tilde{G} = \sum_{j \leq n} q_j w^j$; Parseval on the unit circle gives $\sum |q_j|^2 \leq \sup_{|w|=1} |G - \tilde{G}|^2$, and Cauchy-Schwarz gives $\frac{1}{2} \sum |q_j| \leq \frac{1}{2} \sqrt{n+1} \sup |G - \tilde{G}|$. For the last assertion take $K = \lfloor \delta n \rfloor$: $\log D_{n,K} = O(n)$ and $\log K! = \delta n \log n + O(n)$. $\square$

The bounds (16.3)–(16.4) on the whole generating function and on the avoidance probability are new here; (16.5) and the $\exp(-\delta n \log n)$ form are a second route to theorem 4.2. Boundary placement can change the exact count, but, when all components are long, its normalized influence is smaller than every power of $1/n$; the theorem compares shape-dependent corrections and does not identify an exponentially small expansion for a single sequence.

Remark 16.4 (Agreement of moments is not equality of laws; manuscript 03). At $n = 12$ compare the position and value path lengths $(6, 6)/(6, 6)$ with $(5, 7)/(4, 8)$. Their moments agree through $k = 4$ and first differ at $k = 5$. The oriented avoidance counts are 222822578 and 222822556, the absolute ones 91310696 and 91284560, by manuscript 03’s independent exact tiling routine. “All algebraic orders agree” must not be read as “the finite distributions are identical”.

Lemma 16.5 (A defect-sensitive moment bound; manuscript 03). *Let $\mu_{k,c}$ be the part of μ_k from selected sets with exactly c components, so $d = k - c$. For $1 \leq c \leq k < n/2$,*

$$\mu_{k,c} \leq \frac{\theta^c}{c!} \binom{k-1}{c-1} n^{-d} D_{n,k} \leq \frac{\theta^k}{k!} \left(\frac{k^2}{n}\right)^d D_{n,k}. \quad (16.6)$$

Proof (manuscript 03). There are at most $\frac{n^c}{c!} \binom{k-1}{c-1}$ selected sets with c components and k edges: choose the c initial vertices, order them, and give them a composition of k into c positive edge counts (overlaps are overcounted). Each has probability at most $\theta^c n^c / n^{k+c} \leq \theta^c n^{-k} D_{n,k}$, because $n^{k+c} \geq n^{k+c} (1 - 2k/n)^{2k}$ as in lemma 16.1. Multiplying, and using $n^c \cdot n^{-k} = n^{-d}$, gives the first bound. For the second, $\binom{k-1}{c-1} = \binom{k-1}{d} \leq k^d$, $k!/c! \leq k^d$ and $\theta \geq 1$. $\square$

Theorem 16.6 (All orders, uniformly over forest shapes; manuscript 03). *Fix $r, s \geq 1$, $\theta \in \{1, 2\}$ and $M \geq 0$, and for large n put*

$$K_n = \left\lfloor \frac{(M+4) \log n}{\log \log n} \right\rfloor. \quad (16.7)$$

For every pair of path forests on n vertices with r and s paths, each path having at least K_n vertices,

$$G(1+u) = e^{\theta u} \sum_{J=0}^M \frac{B_J^{(\theta)}(u; r, s)}{n^J} + O_{M,r,s}(n^{-M-1}), \quad |1+u| \leq 1, \quad (16.8)$$

with B_J the polynomials of (5.11). The implied constant does not depend on the individual path lengths.

Proof (manuscript 03, condensed). Tail. By lemmas 16.1 and 16.2 with $|u| \leq 2$, truncating at $k < K_n$ costs at most $(2\theta)^{K_n} D_{n,K_n} / K_n!$. Since $K_n^2/n \rightarrow 0$, $D_{n,K_n} = \exp(O(K_n^2/n))$, and $\log K_n! = (M+4+o(1)) \log n$; so the cost is $o(n^{-M-1})$. Every retained profile has $k < K_n$ edges and is stable in both forests (Corollary 3.2).

High defect. By lemma 16.5, with $k^2/n < 1/2$ for $k < K_n$, the retained profiles of defect $d \geq M+1$ contribute at most $O(n^{-M-1}) \sum_{k \geq 1} (2\theta)^k k^{2M+2}/k! = O_M(n^{-M-1})$, uniformly in u and in the path lengths.

Low defect. For each of the finitely many long profiles β of defect $d \leq M$, expand the normalized kernel $Q_{(m,\beta)}(1/n)$ of (5.3) through order $M-d$. Manuscript 03 proves that the kernel is analytic and bounded by a constant depending only on r, s on $|x| \leq 1/(D_0(1+k)^2)$ for a suitable D_0 , which is the role of Lemma 6.1; Cauchy’s estimate gives a remainder $O((1+k)^{2(M-d)+2} n^{-(M-d-1)})$, summable against the weights $(\theta|u|)^m/m!$ of the dimers.

Completion. Extending each retained dimer sum from $m+K < K_n$ to all $m \geq 0$ adds tails of order $\exp(-K_n \log K_n + O(K_n)) = o(n^{-M-1})$. The completed sums are the Poisson transforms (5.10) of the polynomials (5.7), and collecting the powers of $1/n$ gives (5.11). No step uses the path lengths beyond the stability of the retained profiles. $\square$

Corollary 16.7 (How long is long enough; manuscript 03). *For fixed r, s , the expansion (16.8) holds to every fixed order for every family of forests whose shortest path length $L_{\min}(n)$ satisfies*

$$\frac{L_{\min}(n) \log \log n}{\log n} \rightarrow \infty. \quad (16.9)$$

In particular the paths need not have lengths proportional to n .

Proof (manuscript 03). For each fixed M , (16.9) eventually gives $L_{\min}(n) \geq K_n$. $\square$

Remark 16.8 (Part I's proof gives the same uniformity on every disk). [Added 1 October 2026, batch 74; this observation is the write's, not either manuscript's.] The remark after the proof of [theorem 1.1](#) claims the theorem for forests whose shortest path is a positive multiple of n . Its proof uses the path lengths only to know that every retained profile, of edge degree at most $T_n = \lfloor (2M + 4) \log n / \log \log n \rfloor$, is stable: the moment tail (4.2), the kernel bound Lemma 6.1 and the defect majorant (6.7) hold for arbitrary forests. Read so, it gives [theorem 1.1](#) uniformly on $|u| \leq U$ for all forests whose shortest path has at least T_n vertices, which also covers the condition (16.9). Manuscript 03 states and proves the shape-uniform form, on the unit disk only.

The theorem gives a remainder at each fixed order; it does not assert that the series in $1/n$ converges, and its proof provides no small optimized numerical constant for a given finite n .

17 The correction polynomials as polynomials in the offsets

Proposition 17.1 (Degree in the offsets; manuscript 03). *For every J and θ there is a polynomial in $\mathbb{Q}[r, s, u]$ of total degree at most J in (r, s) whose values at positive integers r, s are $B_J^{(\theta)}(u; r, s)$; it is symmetric in r, s and has degree at most $2J$ in u .*

Proof (manuscript 03, in Part I's notation). In (5.3), the offset r enters $f_{r, \mathbf{b}}(x)$ only through the rising factorials $(r - 1)^{\bar{h}}$, of degree h in r , and each is accompanied by x^h ; so $[x^j]f_{r, \mathbf{b}}$ has degree at most j in r , and likewise for s . Hence $P_{\beta, j} = [x^j]Q_{(m, \beta)}$ has total degree at most j in (r, s) , and (5.11) uses $P_{\beta, J-d}$ with $d \geq 0$. The other assertions are Proposition 7.1. $\square$

Proposition 17.2 (The third avoidance coefficient; manuscript 03). *For all positive integers r, s ,*

$$\begin{aligned} c_1^{(1)} &= r + s - 1, & c_1^{(2)} &= 2(r + s - 2), \\ c_2^{(1)} &= \frac{1}{2}(r^2 - r + s^2 - s), & c_2^{(2)} &= 2(r^2 + rs - 2r + s^2 - 2s), \\ c_3^{(1)} &= \frac{1}{6}(r + s - 1)(r^2 - 4rs + 4r + s^2 + 4s - 6), & & (17.1) \\ c_3^{(2)} &= \frac{2}{3}(2r^3 + 3r^2 + 9rs - 17r + 2s^3 + 3s^2 - 17s + 10). \end{aligned}$$

The first four are Part I's (1.11), (7.3) and (7.5) at $u = -1$; the closed forms of $c_3^{(\theta)}(r, s)$ are new. Manuscript 03 obtains them by evaluating its finite formula symbolically in r, s (the full B_3 for both orientations is in `data/02-fixed-gap-general_pgf_first_three.txt`); this write checked both against Part I's generator at $(r, s) = (1, 1), (1, 3), (2, 3), (3, 3), (2, 5)$. At $r = s = d$, $c_1^{(2)} = 4(d - 1)$ is Tauraso's correction (1.5).

18 The oriented one-path family: proved integrality

Proposition 18.1 (Rational collapse for one path; manuscript 03). *Let the position forest be a single path ($r = 1$), the value forest have s paths, and $\theta = 1$. Then*

$$n! \mu_k = \binom{n-s}{k} (n-k)! \quad (0 \leq k \leq n-s), \quad (18.1)$$

so that $G(1+u) = \frac{1}{n^s} \sum_{k=0}^{n-s} \frac{u^k}{k!} (n-k)^s$, and, for $|u| \leq 2$ and a suitable constant C_s ,

$$G(1+u) = \frac{e^u}{n^s} \sum_{t \geq 0} \frac{\Delta^t p_n(0)}{t!} u^t + O(\exp\{-n \log n + C_s n\}), \quad p_n(m) = (n-m)^s. \quad (18.2)$$

For $s = 1$ the rational factor is $1 - u/n$; for $s = 2$ it is $1 - 2u/n + u^2/(n(n-1))$.

Proof (manuscript 03). Work on the value side. Any k of its $n-s$ edges contract the value set to $n-k$ blocks, and these blocks can be laid along the position path in $(n-k)!$ ways, with no further restriction; this is (18.1). Then use $(n-s)^k/n^k = (n-k)^s/n^s$. Completing the sum to all $k \geq 0$ adds only zero terms through $k = n$ and a factorial tail bounded by a degree- s polynomial times $2^k/k!$, divided by n^s ; the completed sum is e^u times the finite-difference Poisson transform (5.10) of p_n , a polynomial of degree s in m . $\square$

Corollary 18.2 (An integrality subfamily; manuscript 03). *In the oriented case, if $r = 1$ or $s = 1$, then $B_J^{(1)}(u; r, s) \in \mathbb{Z}[u]$ for every J , and every avoidance coefficient $c_J^{(1)}(r, s)$ is an integer.*

Proof (manuscript 03). The finite sum in (18.2) is $\sum_q a_q(n) T_q(u)$, where $p_n(m) = \sum_q a_q(n) m^q$ has integer coefficients and $T_q(u) = e^{-u} \sum_{m \geq 0} u^m m^q / m!$ are the Touchard polynomials, which have integer coefficients; so it is an integer polynomial in n and u of degree s in n . Dividing by $n^s = n^s \prod_{i < s} (1 - i/n)$ and expanding $\prod_{i < s} (1 - i/n)^{-1}$, whose power-series coefficients are integers, gives an expansion in $1/n$ with coefficients in $\mathbb{Z}[u]$. By uniqueness of asymptotic coefficients it is $\sum_J B_J^{(1)}(u; 1, s) n^{-J}$. The case $s = 1$ follows by the symmetry $B_J(u; r, s) = B_J(u; s, r)$. $\square$

This proves integrality only for the oriented one-path subfamily. It says nothing about the case (2, 2) of A189281, whose all-orders integrality remains conditional on Conjecture 11.1 (Proposition 11.2); manuscript 03 likewise warns that the subfamily does not justify extrapolating convergence, termination or integrality to two paths on both sides. At $(r, s) = (1, 1)$ the factor $1 - u/n$ gives $c_J^{(1)}(1, 1) = 1, 1, 0, 0, \dots$, the values found in Remark 7.2.

19 Fixed local probabilities

For fixed j and M , coefficient extraction from (1.7) on a fixed circle gives

$$\mathbb{P}(X = j) = e^{-\theta} \frac{\theta^j}{j!} \left\{ \sum_{J=0}^M \frac{1}{n^J} \sum_h b_{J,h} H_h(j; \theta) + O_j(n^{-M-1}) \right\}, \quad (19.1)$$

with $B_J = \sum_h b_{J,h} u^h$ and the Charlier polynomials (7.6). This is the pointwise form of Corollary 7.3, which is the stronger statement (it controls the summed ℓ^1 error); it asserts no relative-error expansion uniformly for j growing with n . Manuscript 03 makes the first order explicit for the two sequences: from $B_1^{(1)} = -3u$ and $B_1^{(2)} = -6u - 2u^2$ at $(r, s) = (2, 2)$,

$$\mathbb{P}(X_{2,2,n}^{(1)} = j) = \frac{e^{-1}}{j!} \left\{ 1 + \frac{3(1-j)}{n} + O_j(n^{-2}) \right\}, \quad (19.2)$$

$$\mathbb{P}(X_{2,2,n}^{(2)} = j) = e^{-2} \frac{2^j}{j!} \left\{ 1 + \frac{4-j(j+1)/2}{n} + O_j(n^{-2}) \right\}. \quad (19.3)$$

At $j = 0$ these are the first avoidance corrections; at $j = 1, 2$ they show how probability mass is redistributed, which an avoidance count alone does not.

20 The inverse in a coordinate absorbing the constant

For A189281 and A110128 the logarithmic coefficients (9.4) are

Sequence	a_1	a_2	a_3
A189281	37/12	-5/2	1439/360
A110128	49/12	0	4319/360

The first row and a_1 for A110128 are in section 9; a_2 and a_3 for A110128 are added by manuscript 03.

Manuscript 03 inverts (9.3) in coordinates that absorb κ_θ . For large Y put

$$\Lambda = \log Y - \kappa_\theta, \quad N = \frac{\Lambda}{W_0(\Lambda/e)} = \exp(W_0(\Lambda/e) + 1), \quad \Delta = \log N, \quad (20.1)$$

so that $N(\log N - 1) = \Lambda$ exactly. Part I's coordinates (9.6) are the same construction with $\log Y$ in place of Λ ; so $N(Y)$ is Part I's z evaluated at $e^{-\kappa_\theta} Y$.

Proposition 20.1 (First inverse corrections; manuscript 03). *The formal inverse $n_*(Y)$ of (9.3) satisfies*

$$n_*(Y) = N - \frac{1}{2} + \frac{1/8 - a_1}{N\Delta} + \frac{1/24 - a_1/2 - a_2}{N^2\Delta} + O\left(\frac{1}{N^3\Delta}\right), \quad (20.2)$$

and to every fixed order $n_*(Y) = N - \frac{1}{2} + \sum_{j \leq J} \omega_j(\Delta) N^{-j} + O(N^{-J-1}\Delta^{-1})$, with each ω_j rational in Δ and computable from $a_1, \dots, a_j$; for instance

$$\omega_3(\Delta) = -\frac{1}{\Delta} \left\{ \frac{(1/8 - a_1)^2}{\Delta} + \frac{(1/8 - a_1)^2}{2\Delta^2} + \frac{a_1}{4} + a_2 + a_3 - \frac{1}{64} \right\}. \quad (20.3)$$

For the two sequences,

$$n_*(Y) = N - \frac{1}{2} - \frac{71}{24N\Delta} + \frac{1}{N^2\Delta} + \dots, \quad n_*(Y) = N - \frac{1}{2} - \frac{95}{24N\Delta} - \frac{2}{N^2\Delta} + \dots, \quad (20.4)$$

with κ_1 and κ_2 respectively in (20.1).

Proof (manuscript 03). Put $n = N + \delta$. The dominant part $x \log x - x$ has derivative Δ and second derivative $1/N$ at N ; after subtracting Λ the equation becomes $0 = \Delta\delta + \frac{\delta^2}{2N} - \frac{\delta^3}{6N^2} + \dots + \frac{\Delta}{2} + \frac{1}{2} \log(1 + \delta/N) + \sum_j a_j (N + \delta)^{-j}$. The constant term gives $\delta = -\frac{1}{2} + O(1/(N\Delta))$; with $\delta = -\frac{1}{2} + \sum_j \omega_j N^{-j}$ the next two orders read $\Delta\omega_1 + a_1 - \frac{1}{8} = 0$ and $\Delta\omega_2 + \frac{1}{2}a_1 + a_2 - \frac{1}{24} = 0$. At each later order the new coefficient enters only as $\Delta\omega_j$. For a fixed truncation the uncanceled residual is $O(N^{-J-1})$ and the derivative of the smooth carrier is $\Delta + O(1/N)$, so the mean value theorem divides it by a quantity asymptotic to Δ . $\square$

The first coefficient agrees with Part I: $1/8 - a_1 = 1/24 - c_1$ is the coefficient $-(c_1 - 1/24)$ of theorem 9.2, which in Part I's coordinates also carries the terms in κ_θ that are absorbed here. Manuscript 03 interprets n_* as the inverse of the same smooth carrier as Part I, with $X_M(A(n)) = n + O(n^{-M-1}/\log n)$, and brackets the integer threshold $\nu(Y) = \min\{n : A(n) \geq Y\}$ between $\lceil X_M(Y) \mp \epsilon_Y \rceil$, $\epsilon_Y = O(N^{-M-1}/\log N)$, with an existence constant only; this bracket is `p0:thm:staircase` of [12]. It notes that expanding W_0 in logarithms would give the familiar nested logarithmic transseries in Y , and that Λ depends on θ , so one Λ must not be reused for both sequences.

21 Manuscript 03's computations

Manuscript 03's `verify.py` (shipped as `code/02-fixed-gap-verify.py`) checks, with assertions: 90 avoidance counts and 90 complete histograms against brute-force permutations for $n = 0, \dots, 8$ and the offset pairs $(1, 1), (1, 2), (2, 2), (2, 3), (3, 3)$, both orientations; its exact tiling enumerator against A189281 for $n = 0, \dots, 26$ and $n = 40$; 132 stable-profile identities and a nonrealizable zero case; the moment agreement and disagreement of [remark 16.4](#); 50 one-path identities ([18.1](#)); all posted coefficients of both sequences through order ten, and orders 11 and 12; the general first correction and the mean through order three; and that the residual of ([20.4](#)) vanishes through N^{-3} . Its exact enumerator and its coefficient engine are independent of each other and of Part I's programs. At order twelve the engine enumerates the 77 long profiles of defect twelve, 272 of defect at most twelve, with the number of dimers summed exactly.

n	$\varepsilon_2(n)$	$\varepsilon_5(n)$	$\varepsilon_{10}(n)$	$\varepsilon_{12}(n)$
20	$1.2642 \cdot 10^{-4}$	$4.8572 \cdot 10^{-7}$	$-2.3683 \cdot 10^{-10}$	$5.4066 \cdot 10^{-11}$
40	$1.5661 \cdot 10^{-5}$	$6.9763 \cdot 10^{-9}$	$-1.3658 \cdot 10^{-13}$	$4.8917 \cdot 10^{-15}$
80	$1.9541 \cdot 10^{-6}$	$1.0406 \cdot 10^{-10}$	$-6.8418 \cdot 10^{-17}$	$5.2425 \cdot 10^{-19}$
120	$5.7883 \cdot 10^{-7}$	$8.9918 \cdot 10^{-12}$	$-7.9394 \cdot 10^{-19}$	$2.5777 \cdot 10^{-21}$

Table 6: Manuscript 03: the unscaled errors $\varepsilon_M(n) = e \text{A189281}(n)/n! - \sum_{j \leq M} c_j^{(1)}(2, 2)n^{-j}$, at 110-digit working precision, rounded. The values A189281(80) and A189281(120) are b-file inputs [[13](#)], not regenerated. These are uncertified decimals; Part I's [table 2](#) gives certified intervals for the scaled errors at $n = 80, 160, 320$.

[Added 1 October 2026, batch 74: scaled by n^{M+1} , the last two columns are consistent with Part I's higher coefficients: $n^{11}\varepsilon_{10}(n) = -58771$ and -58990 at $n = 80, 120$, against $c_{11}^{(1)}(2, 2) = -59103$; $n^{13}\varepsilon_{12}(n) = 2.882 \cdot 10^6$ and $2.758 \cdot 10^6$, approaching $c_{13}^{(1)}(2, 2) = 2525459$ from above as the next term $c_{14}^{(1)}/n$ with $c_{14}^{(1)} = 26632378$ predicts $(2525459 + 26632378/80 \approx 2.858 \cdot 10^6)$.] The four-term inverse ([20.4](#)) at $Y = \text{A189281}(n)$ misses n by about $2.0566 \cdot 10^{-4}$, $1.9555 \cdot 10^{-5}$, $1.9282 \cdot 10^{-6}$ and $5.0462 \cdot 10^{-7}$ at $n = 20, 40, 80, 120$; the signs are not predicted by the theorem. Manuscript 03 records its run as “Completed in 22.44 seconds” and its README speaks of “tens of seconds”. On this repository's machine the order-twelve run took more than 170 seconds at the intake and 104 seconds at the write, and an order-nine run 72 seconds. The run time depends on the machine and its load.

22 Further questions raised by manuscript 03

Manuscript 03 poses twelve research questions, as proposals only. Six overlap [section 12](#): a certificate for the particular recurrence (Part I's question 2); integrality beyond the one-path case, which asks whether $B_J^{(1)}(u; r, s) \in \mathbb{Z}[u]$ for all r, s and so whether all $c_J^{(1)}(2, 2)$ are integers (question 1, and [corollary 18.2](#) for what is proved); large-order growth (question 4); growing gaps and component counts (question 5); cycles and other sparse constraint graphs (question 6); and a small formal package, starting from the profile identity, the stable polynomial, a verified coefficient checker and the universality bound (question 7). The other six are new here.

1. *A sharper oriented degree bound.* Does $\deg_u B_J^{(1)}(u; r, s) \leq J$ hold at every order, sharpening $2J$? The explicit polynomials through order three show the cancellation; such a bound would constrain the factorial cumulants of X .
2. *Sharp short-path thresholds.* Condition ([16.9](#)) is sufficient, not shown optimal. What is the first shape-dependent term when a shortest path has fixed length, or grows like $c \log n / \log \log n$?

3. *Optimal quantitative universality.* Find the correct constant in the $n \log n$ exponent of [theorem 16.3](#) for given proportions, and the path balances that maximize the difference.
4. *Several gaps simultaneously.* The joint law of violations for several position/value gap pairs, with interacting edge colours and components containing cycles; a first target is two pairs sharing one gap.
5. *Growing local counts and moderate deviations.* Extend [\(19.1\)](#) to a quantified range $j = j(n) \rightarrow \infty$.
6. *Faster generation and certified errors.* Reduce the cost of many orders through profile symmetries, falling-factorial bases or direct Touchard operations, and extract explicit constants from [theorem 16.6](#) to obtain a certified finite-input threshold algorithm (compare Part I's question 8).

OEIS and formalization, as manuscript 03 proposes them. It suggests an OEIS addition stating that the expansions through n^{-10} have a recurrence-independent proof, appending orders 11 and 12, and keeping the recurrences' conjectural status; it submitted nothing, and Part I already has orders 11 to 16. It proposes a formalization in three layers (finite combinatorics; algebra of the stable polynomial and coefficient generation; analysis), with a coefficient checker that verifies equality with the finite formula rather than with a guessed recurrence. It supplies no Lean source, and nothing in Part II is formalized.

The shipped files of manuscript 03. The ten files of manuscript 03 carry the prefix 02-fixed-gap-: its two scripts in `code/`, its six recorded outputs and `requirements.txt` in `data/`, and `sources.md` at the report root, all byte-identical to the delivery. `verify.py` imports its sibling by the delivered name (`from fixed_gap import ...`) and writes unprefixes files into `data/` beside its own directory; run from `code/` it would fail to import, and if made to run it would write beside Part I's files under unprefixes names. The report README gives a copy-based recipe.

A Proof dependencies and archive guide

The main theorem uses the following chain, all proved in the body of the article:

$$\begin{aligned}
 \text{selected-edge tilings} &\implies \text{stable polynomial moments} \\
 &\implies \text{uniform profile expansion} \\
 &\implies \text{finite coefficient formula.}
 \end{aligned}$$

The separate factorial-moment bound justifies discarding the tail and also proves the total-variation universality theorem. Stirling's expansion and the definition of Lambert W are standard external analytic inputs for the inverse section. The recurrence conjectures, the rational-collapse conjecture, and the 2026 holonomicity report are not dependencies of the all-orders theorem.

The archive contains the L^AT_EX source and compiled PDF, the core exact coefficient/enumeration library, a validation program, the rational-certificate program, two optional exploratory programs, and their outputs. The README gives commands to reproduce the computations and compile the article. The symbolic collapse script asserts only its explicitly bounded range of identities; its success is not presented as an infinite proof. [Added 1 October 2026, batch 7302: in the shipped report the compiled PDF is a new build of this text, and the archive's README is replaced by the report README; [section B](#) lists what is shipped.]

B Provenance and editorial notes

[Added 1 October 2026, batch 73O2.]

Source. The report was built from one manuscript, manuscript 60 of batch 73 of ProveIt’s incoming-reports intake: the archive `oeis_path_forests.zip` (main file `article.tex`, a 21-page PDF titled *Unconditional all-orders asymptotics for two OEIS permutation sequences*). It arrived in commit `aa43cc555` and was placed in commit `6e193dd4f`, which deleted the archive from `docs/incoming`; it survives in `aa43cc555`. The manuscript pins no repository revision. It cites the repository only by the path of the Fabius completion audit [9], which it uses as an editorial example and calls “not a dependency”. This report does not continue that audit, and nothing was filed in the Fabius tree. [Added 1 October 2026, batch 74: since batch 74 the report is built from two manuscripts; Part II and the paragraph “Part II (batch 74)” below describe the second.]

What the write changed.

- Every label gained the prefix `spf :`; no label was renamed otherwise and none was removed. The manuscript had 78 labels.
- Added: the editorial note after the status paragraph; Remark 7.2 on the offsets (1, 1) (A000255 and A002464); Remark 9.1 on the repository’s inversion apparatus; bracketed notes on Padhi’s preprint, on integer thresholds and on the shipped files; three bibliography entries [10, 11, 12]; and this section. A separate reciprocal note, Remark 7.4, points to the local law of the companion report on balanced Smirnov words.
- Nothing was removed. Every non-claim of the manuscript is kept: the status paragraph, the Poincaré-only reading of Table 1, the limitation after theorem 4.2, the fixed-offset limitation after the proof of theorem 1.1, the conditional nature of Proposition 11.2, and the statement that the recurrences of A189281 and A110128 are not proved.

Status of the conjecture. Conjecture 11.1 (rational collapse) is a conjecture. It has been checked as an identity of rational functions for $h = 4, \dots, 16$ only, and the integrality of every $c_j^{(1)}(2, 2)$ is conditional on it (Proposition 11.2).

Overloaded letters. The manuscript reuses a few letters; none is renamed here. C is a tiling count $C_{\mathcal{P}}(\mathbf{b})$ in sections 2 and 3, the number of non-dimer components in section 5, a constant $C_{r,s}$ in Lemma 6.1, and the series $C(x)$ in section 9. K is a moment cutoff in Lemma 4.1 and Theorem 4.2 but the edge degree of the non-dimer part of a profile in section 5. d is the defect of a profile, while $d_j(w)$ are the inverse coefficients of section 9. $\mathcal{D}$ is a formal series in section 3 and D the index correction in section 9. In each case the section determines the meaning.

Checks by the intake. The intake reran every shipped program on a copy (all passed; the regenerated outputs equal the shipped ones apart from line endings, a final newline and timings), recounted both sequences for $n \leq 9$ by brute force, compared the two order-ten expansions with the OEIS entries digit for digit, rederived d_1 in theorem 9.2 by hand, and made the (1, 1) checks of Remark 7.2. These checks do not replace an independent review of the proofs.

Part II (batch 74). [Added 1 October 2026, batch 74.] Part II is built from manuscript 03 of batch 74, the archive `ProveIt_Fixed_Gap_Asymptotics.zip` (14 files, a 25-page PDF), which arrived in commit `c664fc2f0` and was placed in commit `b669cff87`. Its pinned commit is `291fbe6bb` [14]. Its article, README, PDF and checksum ledger `SHA256SUMS.txt` (13/13

verified at placement) are not shipped; its other ten files are, byte-identical, with the prefix `02-fixed-gap-`: `code/fixed_gap.py` and `code/verify.py` in `code/`; the six files of its `data/` and `requirements.txt` in `data/`; and `sources.md` at the report root. Those files use the delivery names.

Where the batch-74 merge had to choose. Manuscript 60 stays the base, being the more general in u and the stronger in arithmetic (denominator theorem, order sixteen), in the whole-law statement and in its certificates. Manuscript 03’s re-derivations of Part I’s theorems are recorded as second routes in [table 5](#) and not reprinted; its own results form Part II, printed after Part I’s conclusion so that no existing number changes. Its symbols are renamed into Part I’s notation ([table 4](#)); the devices its new proofs need (the moment bound with $D_{n,k}$, the unit-disk Taylor bound, the defect-sensitive bound) are printed with them. Its coefficients through order twelve are printed only as a confirmation of [table 1](#).

What the batch-74 write changed. It added the part headings, the editorial note after the batch-73O2 one, bracketed notes marked “[Added 1 October 2026, batch 74]” after the proof of [theorem 1.1](#), in questions 1 and 5 of [section 12](#) and in this appendix, Part II ([section 14](#) to [section 22](#)), two bibliography entries, and a table-of-contents line for the appendices. All its labels carry the prefix `fga:`. No existing label was renamed or removed, no existing number changed, and no delivered sentence of manuscript 60 was changed or removed. Manuscript 03’s text is restated in Part I’s notation, with some proofs condensed; none of its statements was strengthened.

References

- [1] OEIS Foundation Inc., *A189281: Number of permutations p of $1, 2, \dots, n$ satisfying $p(i+2) - p(i) \neq 2$* . <https://oeis.org/A189281>. Entry inspected 1 October 2026.
- [2] OEIS Foundation Inc., *A110128: Number of permutations p of $1, 2, \dots, n$ satisfying $|p(i+2) - p(i)| \neq 2$* . <https://oeis.org/A110128>. Entry inspected 1 October 2026.
- [3] G. Spahn and D. Zeilberger, *Counting permutations where the difference between entries located r places apart can never be s (for any given positive integers r and s)*, arXiv:2211.02550 (2022), revised manuscript with postscript. <https://sites.math.rutgers.edu/~zeilberg/mamarim/mamarimPDF/permsV2.pdf>.
- [4] R. Tauraso, *The dinner table problem: the rectangular case*, *Integers* **6** (2006), A11; arXiv:math/0507293. <https://arxiv.org/abs/math/0507293>.
- [5] M. Kauers and C. Koutschan, *Guessing with little data*, *Proceedings of ISSAC 2022*, 83–90; arXiv:2202.07966. <https://arxiv.org/abs/2202.07966>.
- [6] J. S. Padhi, *Solutions to five challenge problems in enumerative and algorithmic combinatorics, with an account of the human–machine methodology employed*, arXiv:2608.11290 (2026), especially Sections 5 and 6. <https://arxiv.org/pdf/2608.11290>. Consulted as a report of current related work; not used as a theorem dependency.
- [7] NIST Digital Library of Mathematical Functions, *Section 5.11: Asymptotic expansions of the gamma function*. <https://dlmf.nist.gov/5.11>. Inspected 1 October 2026.
- [8] NIST Digital Library of Mathematical Functions, *Section 4.13: Lambert W -function*. <https://dlmf.nist.gov/4.13>. Inspected 1 October 2026.

- [9] V. Reshetnikov, *ProveIt*, repository and asymptotic completion audit: `Analysis/FabiusFunction/docs/ASYMPTOTIC_COMPLETION_AUDIT.md`. <https://github.com/VladimirReshetnikov/ProveIt>. Repository material inspected 1 October 2026. Used as project context and an example of separating asymptotic assertions from their verification status.
- [10] [Added in the batch-73O2 write.] OEIS Foundation Inc., *A000255: $a(n) = n a(n-1) + (n-1) a(n-2)$, $a(0) = a(1) = 1$* . <https://oeis.org/A000255>. Entry inspected 1 October 2026.
- [11] [Added in the batch-73O2 write.] OEIS Foundation Inc., *A002464: Hertzprung's problem*, including the Abramson–Moser asymptotic expansion computed by V. Kotesovec. <https://oeis.org/A002464>. Entry inspected 1 October 2026.
- [12] [Added in the batch-73O2 write.] V. Reshetnikov and contributors, *Transseries and Inversion*, canonical volume of the ProveIt series-and-transseries collection. Its source is `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`. Cited here: Part “Inversion: the apparatus for a rapidly growing function” (labels `p0:thm:perturbed-inversion`, `p0:thm:staircase`), chapter on the increasing inverse of Gamma (`p6:thm:gamma`) and chapter “The subfactorial” (`p8:sec:top`). Repository state of placement commit `6e193dd4f`.
- [13] [Added in the batch-74 write, from manuscript 03.] R. Matsuo, V. Kotesovec and C. Koutschan, *Table of A189281 for $n = 0, \dots, 300$* , with the authorship of ranges given in the OEIS entry. <https://oeis.org/A189281/b189281.txt>. Accessed by manuscript 03 on 1 October 2026.
- [14] [Added in the batch-74 write, from manuscript 03.] V. Reshetnikov, *ProveIt*, repository and README, inspected 1 October 2026 at commit `291fbe6bb4477e0cb2131f1e041b12a9abe363fd`. <https://github.com/VladimirReshetnikov/ProveIt>.